CROWD-SOURCED TEXT ANALYSIS: REPRODUCIBLE AND AGILE PRODUCTION OF POLITICAL DATA*

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Abstract

Empirical social science often relies on data that are not observed in the field, but are transformed into quantitative variables by expert researchers who analyze and interpret qualitative raw sources. While generally considered the most valid way to produce data, this expert-driven process is inherently difficult to replicate or to assess on grounds of reliability. Using crowd-sourcing to distribute text for reading and interpretation by massive numbers of non-experts, we generate results comparable to those using experts to read and interpret the same texts, but do so far more quickly and flexibly. Crucially, the data we collect can be reproduced and extended cheaply and transparently, making crowd-sourced datasets intrinsically reproducible. This focuses researchers' attention on the fundamental scientific objective of specifying reliable and replicable methods for collecting the data needed, rather than on the content of any particular dataset. We also show that our approach works straightforwardly with different types of political text, written in different languages. While findings reported here concern text analysis, they have far-reaching implications for expert-generated data in the social sciences.

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Political scientists have made great strides toward greater reproducibility of their findings since the publication of Gary King's influential paper *Replication, Replication* (King 1995). It is now standard practice for good professional journals to insist that authors lodge their data and code in a prominent open access repository. This allows other scholars to replicate and extend published results by reanalyzing the data, rerunning and modifying the code. Replication of an *analysis*, however, sets a far weaker standard than reproducibility of the *data*, which is typically seen as a fundamental principle of the scientific method. Here, we propose a step towards a more comprehensive scientific replication standard in which the mandate is to replicate data production, not just data analysis. This shifts attention from specific datasets as the essential scientific objects of interest, to the *published and reproducible method by which the data were generated.*

We implement this more comprehensive replication standard for the rapidly expanding project of analyzing the content of political texts. Traditionally, a lot of political data is generated by experts applying comprehensive classification schemes to raw sources in a process that, while in principle repeatable, is in practice too costly and time-consuming to reproduce. Widely used examples include:[1] the *Polity* dataset, rating countries on a scale "ranging from -10 (hereditary monarchy) to +10 (consolidated democracy)";[2] the *Comparative Parliamentary Democracy* data with indicators, of the "number of inconclusive bargaining rounds" in government formation and "conflictual" government terminations[3]; the *Comparative Manifesto Project* (CMP), with coded summaries of party manifestos, notably a widely-used left-right score[4]; and the *Policy Agendas*

[1] Other examples of coded data include: expert judgments on party policy positions of party positions (Benoit and Laver 2006; Hooghe et al. 2010; Laver and Hunt 1992); democracy scores from *Freedom House* and corruption rankings from *Transparency International*.
[2] http://www.systemicpeace.org/polity/polity4.htm
[3] http://www.erdda.se/cpd/data_archive.html
[4] https://manifesto-project.wzb.eu/

Project, which codes text from laws, court decisions, political speeches into topics and subtopics (Jones 2013). In addition to the issue of reproducibility, the fixed nature of these schemes and the considerable infrastructure required to implement them discourages change and makes it harder to adapt them to specific needs, as the data are designed to fit general requirements rather than a particular research question.

Here, we demonstrate a method of crowd-sourced text annotation for generating political data that is both *reproducible* in the sense of allowing the data generating process to be quickly, inexpensively, and reliably repeated, and *agile* in the sense of being capable of flexible design according to the needs of a specific research project. The notion of agile research is borrowed from recent approaches to software development, and incorporates not only the flexibility of design, but also the ability to iteratively test, deploy, verify, and if necessary, redesign data generation through feedback in the production process. In what follows, we outline the development of this method to a common measurement problem in political science: locating political parties on policy dimensions using text as data. Despite the lower expertise of crowd workers compared to country experts, properly deployed crowd-sourcing generates results indistinguishable from expert approaches. Data collection can also be *repeated* as often as desired, quickly and with low cost, given the millions of available workers online. Furthermore, our approach is easily tailored to specific research needs, for specific contexts and time periods, in sharp contrast to large "canonical" data generation projects aimed at maximizing generality. For this reason, crowd-sourced data generation represents a paradigm shift for data production and reproducibility in the social sciences. While we apply our particular method for crowd-sourced data production to the analysis of political texts, the core problem of specifying *a reproducible data production process* extends to almost all subfields of political science.

In what follows, we first review the theory and practice of crowd sourcing. We then deploy an experiment in content analysis designed to evaluate crowd sourcing as a method for reliably and validly extracting meaning from political texts, in this case party manifestos. We compare expert and crowd-sourced analyses of the same texts, and assess external validity by comparing crowd-sourced estimates with those generated by completely independent expert surveys. In order to do this we design a method for aggregating judgments about text units of varying complexity, by readers of varying quality,[5] into estimates of latent quantities of interest. To assess the external validity of our results, our core analysis uses crowd workers to estimate party positions on two widely used policy dimensions: "economic" policy (right-left) and "social" policy (liberal-conservative). We then use our method to generate "custom" data on a variable not available in canonical datasets, in this case party policies on immigration. Finally, to demonstrate the truly general applicability of crowd-sourced text annotation to generate data, we test the method in a multi-lingual and technical environment to show that data generation using crowd-sourced text analysis is effective for texts other than party manifestos and works well in different languages.

HARVESTING THE WISDOM OF CROWDS

The intuition behind crowd-sourcing can be traced to Aristotle (Lyon and Pacuit 2013) and later Galton (1907), who noticed that the average of a large number of individual judgments by fair-goers of the weight of an ox is close to the true answer and, importantly, closer to this than the typical individual judgment (for a general introduction see Surowiecki 2004). Crowd-sourcing is now understood to mean using the Internet to distribute a large package of small tasks to a large

[5] In what follows we use the term "reader" to cover a person, whether expert, crowd worker or anyone else, who is evaluating a text unit for meaning.

number of anonymous workers, located around the world and offered small financial rewards per task. The method is widely used for data-processing tasks such as image classification, video annotation, data entry, optical character recognition, translation, recommendation, and proofreading. Crowd-sourcing has emerged as a paradigm for applying human intelligence to problem-solving on a massive scale, especially for problems involving the nuances of language or other interpretative tasks where humans excel but machines perform poorly.

Increasingly, crowd-sourcing has also become a tool for social scientific research (Bohannon 2011). In sharp contrast to our own approach, most applications use crowds as a cheap alternative to traditional subjects for experimental studies (e.g. Lawson et al. 2010; Horton et al. 2011; Paolacci et al. 2010; Mason and Suri 2012). Using subjects in the crowd to populate experimental or survey panels raises obvious questions about external validity, addressed by studies in political science (Berinsky et al. 2012), economics (Horton et al. 2011) and general decision theory and behavior (Paolacci et al. 2010; Goodman et al. 2013; Chandler et al. 2014). Our method for using workers in the crowd to label *external* stimuli differs fundamentally from such applications. We do not care at all about whether our crowd workers represent any target population, as long as different workers, on average, make the same judgments when faced with the same information. In this sense our method, unlike online experiments and surveys, is a canonical use of crowd-sourcing as described by Galton.[6]

All data production by humans requires expertise, and several empirical studies have found that data created by domain experts can be matched, and sometimes improved at much lower cost, by aggregating judgments of non-experts (Alonso and Mizzaro 2009; Hsueh et al. 2009; Snow et al. 2008; Alonso and Baeza-Yates 2011; Carpenter 2008; Ipeirotis et al. 2013). Provided crowd workers are not systematically biased in relation to the "true" value of the latent

[6] We are interested in the weight of the ox, not in how different people judge the weight of the ox.

quantity of interest, and it is important to check for such bias, the central tendency of even erratic workers will converge on this true value as the number of workers increases. Because experts are axiomatically in short supply while members of the crowd are not, crowd-sourced solutions also offer a straightforward *and scalable* way to address reliability in a manner that expert solutions cannot. To improve confidence, simply employ more crowd workers. Because data production is broken down into many simple specific tasks, each performed by many different exchangeable workers, it tends to wash out biases that might affect a single worker, while also making it possible to estimate and correct for worker-specific effects using the type of scaling model we employ below.

Crowd-sourced data generation inherently requires a method for aggregating many small pieces of information into valid measures of our quantities of interest.[7] Complex calibration models have been used to correct for worker errors on particular difficult tasks, but the most important lesson from this work is that increasing the number of workers reduces error (Snow et al. 2008). Addressing statistical issues of "redundant" coding, Sheng et al. (2008) and Ipeirotis et al. (2010) show that repeated coding can improve the quality of data as a function of the individual qualities and number of workers, particularly when workers are imperfect and labeling categories are "noisy." Ideally, we would benchmark crowd workers against a "gold standard," but such benchmarks are not always available, so scholars have turned to Bayesian scaling models borrowed from item-response theory (IRT), to aggregate information while simultaneously assessing worker quality (e.g. Carpenter 2008; Raykar et al. 2010). Welinder and Perona (2010) develop a classifier that integrates data difficulty and worker characteristics, while Welinder et al. (2010) develop a unifying model of the characteristics of both data and workers,

[7] Of course aggregation issues are no less important when combining any multiple judgments, including those of experts. Procedures for aggregating non-expert judgments may influence both the quality of data and convergence on some underlying "truth", or trusted expert judgment. For an overview, see Quoc Viet Hung et al. (2013).

such as competence, expertise and bias. A similar approach is applied to rater evaluation in Cao et al. (2010) where, using a Bayesian hierarchical model, raters' judgments are modeled as a function of a latent item trait, and rater characteristics such as bias, discrimination, and measurement error. We build on this work below, applying both a simple averaging method and a Bayesian scaling model that estimates latent policy positions while generating diagnostics on worker quality and sentence difficulty parameters. We find that estimates generated by our more complex model match simple averaging very closely.

A METHOD FOR REPLICABLE CODING OF POLITICAL TEXT

We apply our crowd-sourcing method to one of the most wide-ranging research programs in political science, the analysis of political text, and in particular text processing by *human* analysts that is designed to extract meaning systematically from some text corpus, and from this to generate valid and reliable data. This is related to, but quite distinct from, spectacular recent advances in *automated* text analysis that in theory scale up to unlimited volumes of political text (Grimmer and Stewart 2013). Many automated methods involve *supervised machine learning* and depend on labeled training data. Our method is directly relevant to this enterprise, offering a quick, effective and above all *reproducible* way to generate labeled training data. Other, *unsupervised*, methods intrinsically require *a posteriori* human interpretation that may be haphazard and is potentially biased.[8]

Our argument here speaks directly to more traditional content analysis within the social sciences, which is concerned with problems that automated text analysis cannot yet address. This involves the "reading" of text by real humans who interpret it for meaning. These interpretations,

[8] This human interpretation can be reproduced by workers in the crowd, though this is not at all our focus in this paper.

if systematic, may be classified and summarized using numbers, but the underlying human interpretation is fundamentally qualitative. Crudely, human analysts are employed to engage in *natural language processing* (NLP) which seeks to extract "meaning" embedded in the syntax of language, treating a text as more than a bag of words. NLP is another remarkable growth area, though it addresses a fundamentally difficult problem and fully automated NLP still has a long way to go. Traditional human experts in the field of inquiry are of course highly sophisticated natural language processors, finely tuned to particular contexts. The core problem is that they are in very short supply. This means that text processing by human experts simply does not scale to the huge volumes of text that are now available. This in turn generates an inherent difficulty in meeting the more comprehensive scientific replication standard to which we aspire. Crowd-sourced text analysis offers a compelling solution to this problem. Human workers in the crowd can be seen, perhaps rudely, as generic and very widely available "biological" natural language processors. Our task in this paper is now clear. Design a system for employing generic workers in the crowd to analyze text for meaning in a way that is as reliable and valid as if we had used finely tuned experts to do the same job.

By far the best known research program in political science that relies on expert human readers is the long-running *Manifesto Project* (MP). This project has analyzed nearly 4,000 manifestos issued since 1945 by nearly 1,000 parties in more than 50 countries, using experts who are country specialists to label sentences in each text in their original languages. A single expert assigns every sentence in every manifesto to a single category in a 56-category scheme devised by the project in the mid-1980s (Budge et al. 1987; Laver and Budge 1992; Klingemann et al. 1994; Budge et al. 2001; Klingemann et al. 2006).[9] This has resulted in a widely-used "canonical" dataset that, given the monumental coordinated effort of very many experts over 30

[9] https://manifesto-project.wzb.eu/

years, is unlikely ever to be re-collected from scratch and in this sense is unlikely to be replicated. Despite low levels of inter-expert reliability found in experiments using the MP's coding scheme (Mikhaylov et al. 2012), a proposal to re-process the entire manifesto corpus many times, using many independent experts, is in practice a non-starter. Large canonical datasets such as this, therefore, tend not to satisfy the deeper standard of reproducible research that requires the transparent repeatability of data generation. This deeper replication standard can however be satisfied with the crowd-sourced method we now describe.

A simple coding scheme for economic and social policy

We assess the potential for crowd-sourced text analysis using an experiment in which we serve up an identical set of documents, and an identical set of text processing tasks, to both a small set of experts (political science faculty and graduate students) and a large and heterogeneous set of crowd workers located around the world. To do this, we need a simple scheme for labeling political text that can be used reliably by workers in the crowd. Our scheme first asks readers to classify each sentence in a document as referring to economic policy (left or right), to social policy (liberal or conservative), or to neither. Substantively, these two policy dimensions have been shown to offer an efficient representation of party positions in many countries.[10] They also correspond to dimensions covered by a series of expert surveys (Benoit and Laver 2006; Hooghe et al. 2010; Laver and Hunt 1992), allowing validation of estimates we derive against widely used independent estimates of the same quantities. If a sentence was classified as economic policy, we then ask readers to rate it on a five-point scale from very left to very right; those

[10] See Chapter 5 of Benoit and Laver (2006) for an extensive empirical review of this for a wide range of contemporary democracies.

classified as social policy were rated on a five-point scale from liberal to conservative. Figure 1 shows this scheme.[11]

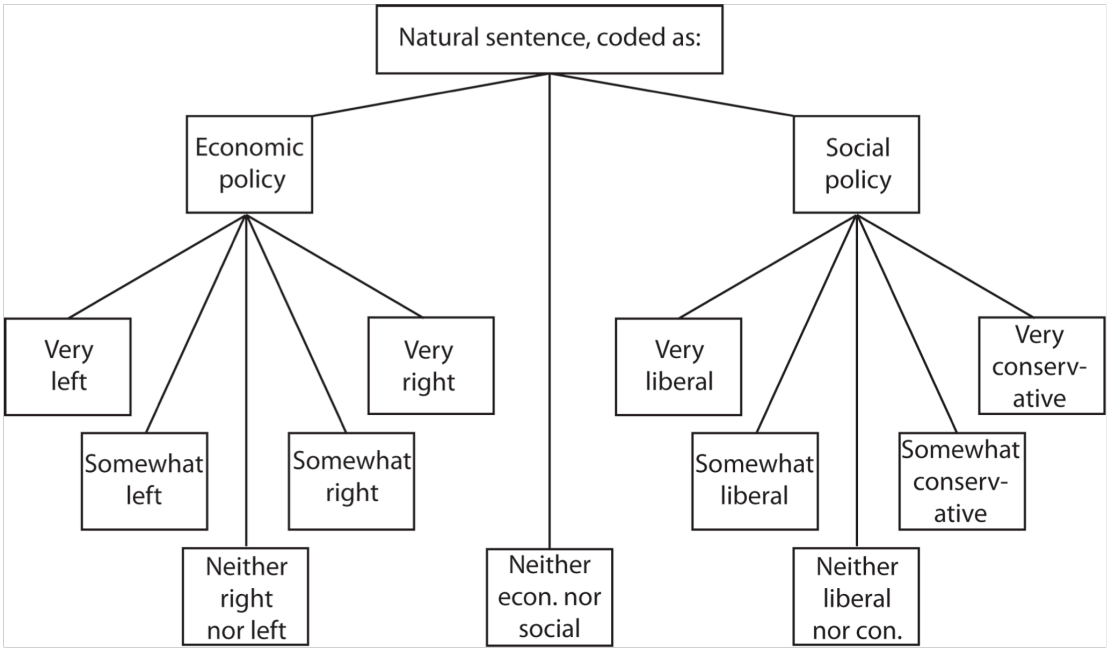


Figure 1: Hierarchical coding scheme for two policy domains with ordinal positioning.

We did not use the MP's 56-category classification scheme, for two main reasons. The first is methodological: complexity of the MP scheme and uncertain boundaries between many of its categories were major sources of unreliability when multiple experts applied this scheme to the same documents (Mikhaylov et al. 2012). The second is practical: it is impossible to write clear and precise instructions, to be understood reliably by a diverse, globally distributed, set of workers in the crowd, for using a detailed and complex 56-category scheme quintessentially designed for highly trained experts. This highlights an important trade-off. There may be data production tasks that cannot feasibly be explained in clear and simple terms, sophisticated instructions that can only be understood and implemented by highly trained experts.

[11] Our instructions—fully detailed in the supplementary materials (section 6)—were identical for both expert and non-experts, defining the economic left-right and social liberal-conservative policy dimensions we estimate and providing examples of labeled sentences.

Sophisticated instructions are designed for a more limited pool of experts who can understand and implement them and, for this reason, imply less scalable and replicable data production. The striking alternative now made available by crowd-sourcing is to break down complicated data production tasks into simple small jobs, as happens when complex consumer products are manufactured on factory production lines. Moreover, even complex text processing tasks involving numerous dimensions can be broken down into separate exercises, as asking the crowd to generate one type of data in no way precludes other tasks from being deployed, concurrently or later, to gather other types of data. Over and above this, the simple scheme in Figure 1 is also motivated by the observation that most scholars using manifesto data actually seek simple solutions, typically estimates of positions on a few general policy dimensions; they do not need estimates of these positions in a 56-dimensional space.

Text corpus

While we extend this in work we discuss below, our baseline text corpus comprises 18,263 natural sentences from British Conservative, Labour and Liberal Democrat manifestos for the six general elections held between 1987 and 2010. These texts were chosen for two main reasons. First, for systematic external validation, there are diverse independent estimates of British party positions for this period, from contemporary expert surveys (Laver and Hunt 1992; Laver 1998; Benoit 2005, 2010) as well as MP expert codings of the same texts. Second, there are well-documented substantive shifts in party positions during this period, notably the sharp shift of Labour towards the center between 1987 and 1997. The ability of crowd workers to pick up this move is a good test of external validity.

In designing the breakdown and presentation of the text processing tasks given to both experts and the crowd, we made a series of detailed operational decisions based on substantial

testing and adaptation (reviewed in the Appendix). In summary, we used natural sentences as our fundamental text unit. Recognizing that most crowd workers dip into and out of our jobs and would not stay online to code entire documents, we served target sentences from the corpus in a random sequence, set in a two-sentence context on either side of the target sentence, without identifying the text from which the sentence was drawn. Our coding experiments showed that these decisions resulted in estimates that did not significantly differ from those generated by the classical approach of reading entire documents from beginning to end.

SCALING DOCUMENT POLICY POSITIONS FROM CODED SENTENCES

Our aim is to estimate the policy positions of entire documents: not the code value of any single sentence, but some aggregation of these values into an estimate of each document's position on some meaningful policy scale while allowing for reader, sentence, and domain effects. One option is simple averaging: identify all economic scores assigned to sentences in a document by all readers, average these, and use this as an estimate of the economic policy position of a document. Mathematical and behavioral studies on aggregations of individual judgments imply that simpler methods often perform as well as more complicated ones, and often more robustly (e.g. Ariely et al. 2000; Clemen and Winkler 1999). Simple averaging of individual judgments is the benchmark when there is no additional information on the quality of individual coders (Lyon and Pacuit 2013; Armstrong 2001; Turner et al. 2013). However, this does not permit direct estimation of misclassification tendencies by readers who for example fail to identify economic or social policy "correctly," or of reader-specific effects in the use of positional scales.

An alternative is to model each sentence as containing information about the document, and then scale these using a measurement model. We propose a model based on item response theory (IRT), which accounts for both individual reader effects and the strong possibility that

some sentences are intrinsically harder to interpret. This approach has antecedents in psychometric methods (e.g. Baker and Kim 2004; Fox 2010; Hambleton et al. 1991; Lord 1980), and has been used to aggregate crowd ratings (e.g. Ipeirotis et al. 2010; Welinder et al. 2010; Welinder and Perona 2010; Whitehill et al. 2009).

We model each sentence, j, as a vector of parameters, θ_{jd}, which corresponds to sentence attributes on each of four latent dimensions, d. In our application, these dimensions are: latent *domain propensity* of a sentence to be labeled economic (1) and social (2) versus none; latent *position* of the sentence on economic (3) and social (4) dimensions. Individual readers i have potential *biases* in each of these dimensions, manifested when classifying sentences as "economic" or "social", and when assigning positions on economic and social policy scales. Finally, readers have four *sensitivities*, corresponding to their relative responsiveness to changes in the latent sentence attributes in each dimension. Thus, the latent coding of sentence j by reader i on dimension d is:

$$\mu^*_{ijd} = \chi_{id}(\theta_{jd} + \psi_{id}) \qquad (1)$$

where the χ_{id} indicate relative *responsiveness* of readers to changes in latent sentence attributes θ_{jd}, and the ψ_{id} indicate relative *biases* towards labeling sentences as economic or social ($d = 1,2$), and rating economic and social sentences as right rather than left ($d = 3,4$).

We cannot observe readers' behavior on these dimensions directly. We therefore model their responses to the choice of label between economic, social and "neither" domains using a multinomial logit given μ^*_{ij1} and μ^*_{ij2}. We model their choice of scale position as an ordinal logit depending on μ^*_{ij3} if they label the sentence as economic and on μ^*_{ij4} if they label the sentence as

social.[12] This results in the following model for the eleven possible combinations of labels and scales that a reader can give a sentence:[13]

$$p(none) = \left(\frac{1}{1 + \exp(\mu^*_{ij1}) + \exp(\mu^*_{ij2})}\right)$$

$$p(econ; scale) = \left(\frac{\exp(\mu^*_{ij1})}{1 + \exp(\mu^*_{ij1}) + \exp(\mu^*_{ij2})}\right)\left(logit^{-1}(\xi_{scale} - \mu^*_{ij3}) - logit^{-1}(\xi_{scale-1} - \mu^*_{ij3})\right)$$

$$p(soc; scale) = \left(\frac{\exp(\mu^*_{ij2})}{1 + \exp(\mu^*_{ij1}) + \exp(\mu^*_{ij2})}\right)\left(logit^{-1}(\xi_{scale} - \mu^*_{ij4}) - logit^{-1}(\xi_{scale-1} - \mu^*_{ij4})\right)$$

The primary quantities of interest are not sentence level attributes, θ_{jd}, but rather aggregates of these for entire documents, represented by the $\bar{\theta}_{k,d}$ for each document *k* on each dimension *d*. Where ϵ_{jd} are distributed normally with mean zero and standard deviation σ_d, we model these latent sentence level attributes θ_{jd} hierarchically in terms of corresponding latent document level attributes:

$$\theta_{jd} = \bar{\theta}_{k(j),d} + \epsilon_{jd}$$

As at the sentence level, two of these (*d=1,2*) correspond to the overall frequency (importance) of economic and social dimensions relative to other topics, the remaining two (*d=3,4*) correspond to aggregate left-right positions of documents on economic and social dimensions.

This model enables us to generate estimates of not only our quantities of interest for the document-level policy positions, but also a variety of reader- and sentence- level diagnostics

[12] By treating these as independent, and using the logit, we are assuming independence between the choices and between the social and economic dimensions (IIA). It is not possible to identify a more general model that relaxes these assumptions without asking additional questions of readers.

[13] Each policy domain has five *scale* points, and the model assumes proportional odds of being in each higher scale category in response to the sentence's latent policy positions θ_3 and θ_4 and the coder's sensitivities to this association. The cutpoints ξ for ordinal scale responses are constrained to be symmetric around zero and to have the same cutoffs in both social and economic dimensions, so that the latent scales are directly comparable to one another and to the raw scales. Thus, $\xi_2 = \infty$, $\xi_1 = -\xi_{-2}$, $\xi_0 = -\xi_{-1}$, and $\xi_{-3} = -\infty$.

concerning reader agreement and the "difficulty" of domain and positional coding for individual sentences. Simulating from the posterior also makes it straightforward to estimates Bayesian credible intervals indicating our uncertainty over document-level policy estimates.[14]

Posterior means of the document level $\bar{\theta}_{kd}$ correlate very highly with those produced by the simple averaging methods discussed earlier: 0.95 and above, as we report below. It is therefore possible to use averaging methods to summarize results in a simple and intuitive way that is also invariant to shifts in mean document scores that might be generated by adding new documents to the coded corpus. The value of our scaling model is to estimate reader and sentence fixed effects, and correct for these if necessary. While this model is adapted to our particular classification scheme, it is general in the sense that nearly all attempts to measure policy in specific documents will combine domain classification with positional coding.

BENCHMARKING A CROWD OF EXPERTS

Our core objective is to compare estimates generated by workers in the crowd with analogous estimates generated by experts. Since readers of all types will likely disagree over the meaning of particular sentences, an important benchmark for our comparison of expert and crowd-sourced text coding concerns levels of disagreement between experts. The first stage of our empirical work therefore employed multiple (four to six)[15] experts to independently code each of the 18,263 sentences in our 18-document text corpus, using the scheme described above. The entire corpus was processed twice by our experts. First, sentences were served in their natural sequence in each manifesto, to mimic classical expert content analysis. Second, about a year later,

[14] We estimate the model by MCMC using the JAGS software, and provide the code, convergence diagnostics, and other details of our estimations in section 2 of the supplementary materials.

[15] Three of the authors of this paper, plus three senior PhD students in Politics from XXX University processed the six manifestos from 1987 and 1997. One author of this paper and four XXX PhD students processed the other 12 manifestos.

sentences were processed in random order, to mimic the system we use for serving sentences to crowd workers. Sentences were uploaded to a custom-built, web-based platform that displayed sentences in context and made it easy for experts to process a sentence with a few mouse clicks. In all, we harvested over 123,000 expert evaluations of manifesto sentences, about seven per sentence. Table 1 provides details of the 18 texts, with statistics on the overall and mean numbers of evaluations, for both stages of expert processing as well as the crowd processing we report below.

Manifesto	Total sentences in manifesto	Mean expert evaluations: natural sequence	Mean expert evaluations: random sequence	Total expert evaluations	Mean crowd evaluations	Total crowd evaluations
Con 1987	1,015	6.0	2.4	7,920	44	36,594
LD 1987	878	6.0	2.3	6,795	22	24,842
Lab 1987	455	6.0	2.3	3,500	20	11,087
Con 1992	1,731	5.0	2.4	11,715	6	28,949
LD 1992	884	5.0	2.4	6,013	6	20,880
Lab 1992	661	5.0	2.3	4,449	6	23,328
Con 1997	1,171	6.0	2.3	9,107	20	11,136
LD 1997	873	6.0	2.4	6,847	20	5,627
Lab 1997	1,052	6.0	2.3	8,201	20	4,247
Con 2001	748	5.0	2.3	5,029	5	3,796
LD 2001	1,178	5.0	2.4	7,996	5	5,987
Lab 2001	1,752	5.0	2.4	11,861	5	8,856
Con 2005	414	5.0	2.3	2,793	5	2,128
LD 2005	821	4.1	2.3	4,841	5	4,173
Lab 2005	1,186	4.0	2.4	6,881	5	6,021
Con 2010	1,240	4.0	2.3	7,142	5	6,269
LD 2010	855	4.0	2.4	4,934	5	4,344
Lab 2010	1,349	4.0	2.3	7,768	5	6,843
Total	18,263	91,400	32,392	123,792		215,107

Table 1. Texts and sentences coded: 18 British party manifestos

External validity of expert evaluations

Figure 2 plots two sets of estimates of positions of the 18 manifestos on economic and social policy: one generated by experts processing sentences in natural sequence (*x*-axis); the other generated by completely independent expert surveys (*y*-axis).[16] Linear regression lines summarizing these plots show that expert text processing predicts independent survey measures very well for economic policy (R= 0.91), somewhat less well for the noisier dimension of social policy (R=0.81). To test whether coding sentences in their natural sequence affected results, our experts also processed the entire text corpus taking sentences in random order. Comparing estimates from sequential and random-order sentence processing, we found almost identical results, with correlations of 0.98 between scales.[17] Moving from "classical" expert content analysis to having experts process sentences served at random from anonymized texts makes no substantive difference to point estimates of manifesto positions. This reinforces our decision to use the much more scalable random sentence sequencing in the crowd-sourcing method we specify.

[16] These were: Laver and Hunt (1992); Laver (1998) for 1997; Benoit and Laver (2006) for 2001; Benoit (2005, 2010) for 2005 and 2010.

[17] Details provided in supplementary materials, section 5.

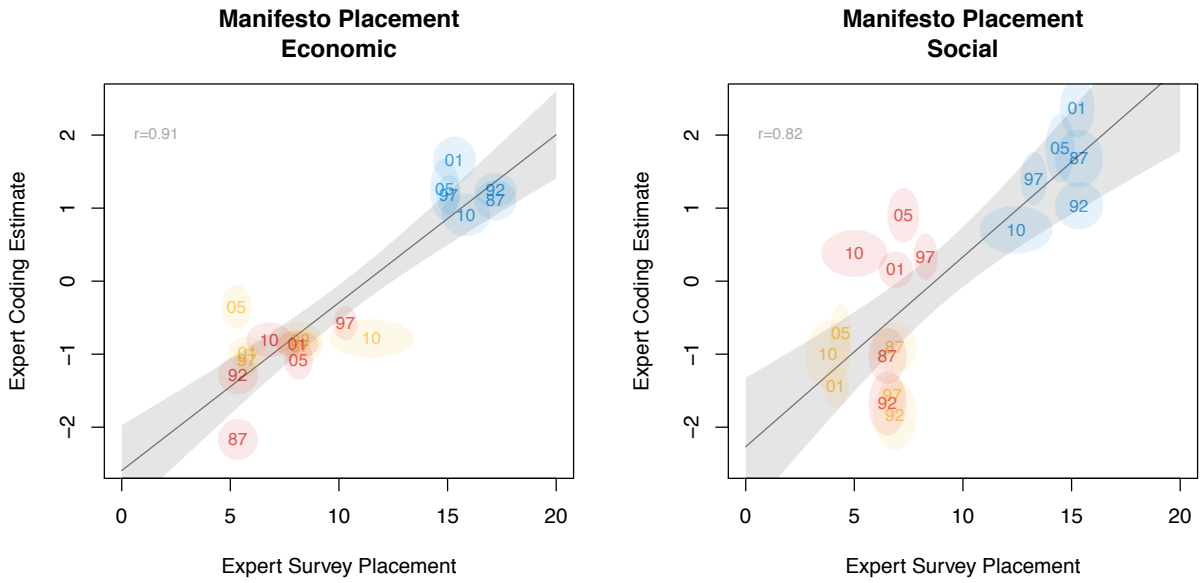


Figure 2. British party positions on economic and social policy 1987 – 2010; sequential expert text processing (vertical axis) and independent expert surveys (horizontal).
(Labour red, Conservatives blue, Liberal Democrats yellow, labeled by last two digits of year)

Internal reliability of expert coding

Agreement between experts

As might be expected, agreement between our experts was far from perfect. Table 2 classifies each of the 5,444 sentences in the 1987 and 1997 manifestos, all of which were processed by the same six experts. It shows how many experts agreed the sentence referred to economic, or social, policy. If experts are in perfect agreement on the policy content of each sentence, either all six label each sentence as dealing with economic (or social) policy, or none do. The first data column of the table shows a total of 4,125 sentences which all experts agree have no social policy content. Of these, there are 1,193 sentences all experts also agree have no economic policy content, and 527 that all experts agree do have economic policy content. The experts disagree about the remaining 2,405 sentences: some but not all experts label these as having economic policy content.

Experts Assigning Economic Domain	*Experts Assigning Social Policy Domain*							
	0	1	2	3	4	5	6	Total
0	**1,193**	196	67	59	114	190	**170**	1,989
1	326	93	19	11	9	19	-	477
2	371	92	15	15	5	-	-	498
3	421	117	12	7	-	-	-	557
4	723	68	10	-	-	-	-	801
5	564	31	-	-	-	-	-	595
6	**527**	-	-	-	-	-	-	527
Total	4,125	597	123	92	128	209	170	5,444

Table 2: Domain classification matrix for 1987 and 1997 manifestos: frequency with which sentences were assigned by six experts to economic and policy domains. (Shaded boxes: perfect agreement between experts.)

The shaded boxes show sentences for which the six experts were in unanimous agreement – on economic policy, social policy, or neither. There was unanimous expert agreement on about 35 percent of the labeled sentences. For about 65 percent of sentences, there was disagreement, even about the policy area, among trained experts of the type usually used to analyze political texts.

Scale reliability

Despite substantial disagreement among experts about individual sentences, we saw above that we can derive externally valid estimates of party policy positions if we aggregate the judgments of all experts on all sentences in a given document. This happens because, while each expert judgment on each sentence is a noisy realization of some underlying signal about policy content, the expert judgments taken as a whole scale nicely – in the sense that in aggregate they are all capturing information about the same underlying quantity. Table 3 shows this, reporting a scale reliability analysis for economic policy positions of the 1987 and 1997 manifestos, derived by treating economic policy scores for each sentence allocated by each of the six expert coders as six sets of independent estimates of economic policy positions.

Item	*N*	*Sign*	*Item-scale correlation*	*Item-rest correlation*	*Cronbach's alpha*
Expert 1	2,256	+	0.89	0.76	0.95
Expert 2	2,137	+	0.89	0.76	0.94
Expert 3	1,030	+	0.87	0.74	0.94
Expert 4	1,627	+	0.89	0.75	0.95
Expert 5	1,979	+	0.89	0.77	0.95
Expert 6	667	+	0.89	0.81	0.93
Overall					0.95

Table 3. Inter-expert scale reliability analysis for the economic policy, generated by aggregating all expert scores for sentences judged to have economic policy content.

Despite the high variance in individual sentence labels we saw in Table 2, overall scale reliability, measured by a Cronbach's alpha of 0.95, is "excellent" by any conventional standard.[18] We can therefore apply our model to aggregate the noisy information contained in the combined set of expert judgment at the sentence level to produce coherent estimates of policy positions at the document level. This is the essence of crowd-sourcing. It shows that our experts are really a small crowd.

DEPLOYING CROWD-SOURCED TEXT CODING

CrowdFlower: a crowd-sourcing platform with multiple channels

Many online platforms now distribute crowd-sourced micro-tasks (Human Intelligence Tasks or "HITs") via the Internet. The best known is Amazon's Mechanical Turk (MT), an online marketplace for serving HITs to workers in the crowd. Workers must often pass a pre-task qualification test, and maintain a certain quality score from validated tasks that determines their status and qualification for future jobs. Rather than relying on a single crowdsourcing platform,

[18] Conventionally, an alpha of 0.70 is considered "acceptable". Nearly identical results for social policy are available in supplementary materials (section 1d). Note that we use Cronbach's alpha as a measure of scale reliability across readers, as opposed to a measure of inter-reader agreement (in which case we would have used Krippendorf's alpha).

we used CrowdFlower, a service that consolidates access to dozens of crowdsourcing channels.[19] This is a viable option because CrowdFlower not only offers an interface for designing templates and uploading tasks but, crucially, also maintains a common training and qualification system for potential workers from any channel before they can qualify for tasks, as well as cross-channel quality control while tasks are being completed.

Quality control

Excellent quality assurance is critical to all reliable and valid data production. Given the natural economic motivation of workers in the crowd to finish as many jobs in as short a time as possible, it is both tempting and easy for workers to submit bad or faked data. Workers who do this are called "spammers". Given the open nature of the platform, it is vital to prevent them from participating in a job, using careful screening and quality control (e.g. Kapelner and Chandler 2010; Nowak and Rger 2010; Eickhoff and de Vries 2012; Berinsky et al. forthcoming). Conway used coding experiments to assess three increasingly strict screening tests for workers in the crowd. (Conway 2013).[20] Two findings directly inform our design. First, using a screening or qualification test *substantially* improves the quality of results; a well-designed test can screen out spammers and bad workers who otherwise tend to exploit the job. Second, once a suitable test is in place, increasing its difficulty *does not* improve results. It is vital to have a filter on the front end to keep out spammers and bad workers, but a tougher filter does not necessarily lead to better workers.

[19] See http://www.crowdflower.com.

[20] There was a baseline test with no filter, a "low-threshold" filter where workers had to correctly code 4/6 sentences correctly, and a "high-threshold" filter that required 5/6 correct labels. A "correct" label means the sentence is labeled as having the same policy domain as that provided by a majority of expert coders. The intuition here is that tough tests also tend to scare away good workers.

The primary quality control system used by CrowdFlower relies on completion of "gold" HITs: tasks with unambiguous correct answers specified in advance.[21] Correctly performing "gold" tasks, which are both used in qualification tests and randomly sprinkled through the job, is used to monitor worker quality and block spammers and bad workers. We specified our own set of gold HITs as sentences for which there was unanimous expert agreement on both policy area (economic, social or neither), and policy direction (left or right, liberal or conservative), and seeded each job with the recommended proportion of about 10% "gold" sentences. We therefore used "natural" gold sentences occurring in our text corpus, but could also have used "artificial" gold sentences, manufactured to represent archetypical economic or social policy statements. We also used a special type of gold sentences called "screeners", (Berinsky et al. forthcoming). These contained an exact instruction on how to label the sentence,[22] set in a natural two-sentence context, and are designed to ensure coders pay attention throughout the coding process.

Specifying gold sentences in this way, we implemented a two-stage process of quality control. First, workers were only allowed into the job if they correctly completed 8 out of 10 gold tasks in a qualification test.[23] Once workers are on the job and have seen at least four more gold sentences, they are given a "trust" score, which is simply the proportion of correctly labeled gold. If workers get too many gold HITs wrong, their trust level goes down. They are ejected from the job if their trust score falls below 0.8. The current trust score of a worker is recorded with each HIT, and can be used to weight the contribution of the relevant piece of information to some aggregate estimate. Our tests showed this weighting made no substantial difference, however, mainly because trust scores all tended to range in a tight interval around a mean of

[21] For CrowdFlower's formal definition of gold, see http://crowdflower.com/docs/gold.
[22] For example, "Please code this sentence as having economic policy content with a score of very right."
[23] Workers giving wrong labels to gold questions are given a short explanation of why they are wrong.

0.84.[24] *Many* more potential HITs than we use here were rejected as "untrusted", because the workers did not pass the qualification test, or because their trust score subsequently fell below the critical threshold. Workers are not paid for rejected HITs, giving them a strong incentive to perform tasks carefully, as they do not know which of these have been designated as gold for quality assurance. We have no hesitation in concluding that a system of thorough and continuous monitoring of worker quality is necessary for reliable and valid crowd sourced text analysis.

Deployment

We set up an interface on CrowdFlower that was nearly identical to our custom-designed expert web system and deployed this in two stages. First, we over-sampled all sentences in the 1987 and 1997 manifestos, because we wanted to determine the number of judgments per sentence needed to derive stable estimates of our quantities of interest. We served up sentences from the 1987 and 1997 manifestos until we obtained a minimum of 20 judgments per sentence. After analyzing the results to determine that our estimates of document scale positions converged on stable values once we had five judgments per sentence—in results we report below—we served the remaining manifestos until we reached five judgments per sentence. In all, we gathered 215,107 judgments by crowd workers of the 18,263 sentences in our 18 manifestos, employing a total of 1,488 different workers from 49 different countries. About 28 percent of these came from the US, 15 percent from the UK, 11 percent from India, and 5 percent each from Spain, Estonia, and Germany. The average worker processed about 145 sentences; most processed between 10 and 70 sentences, 44 workers processed over 1,000 sentences, and four processed over 5,000. [25]

[24] Our supplementary materials (section 4) report the distribution of trust scores from the complete set of crowd-codings by country of the worker and channel, in addition to results that scale the manifesto aggregate policy scores by the trust scores of the workers.

[25] Our final crowd-coded dataset was generated by deploying through a total of 26 CrowdFlower channels. The most common was Neodev (Neobux) (40%), followed by Mechanical Turk (18%), Bitcoinget (15%), Clixsense (13%),

CROWD-SOURCED ESTIMATES OF PARTY POLICY POSITIONS

Figure 3 plots crowd-sourced estimates of the economic and social policy positions of British party manifestos against estimates generated from analogous expert text processing.[26] The very high correlations of aggregate policy measures generated by crowd workers and experts suggest both are measuring the same latent quantities. Substantively, Figure 3 also shows that crowd workers identified the sharp rightwards shift of Labour between 1987 and 1997 on both economic and social policy, a shift identified by expert text processing and independent expert surveys. The standard errors of crowd-sourced estimates are higher for social than for economic policy, reflecting both the smaller number of manifesto sentences devoted to social policy and higher coder disagreement over the application of this policy domain.[27] Nonetheless Figure 3 summarizes our evidence that the crowd-sourced estimates of party policy positions can be used as substitutes for the expert estimates, which is our main concern in this paper.

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and Prodege (Swagbucks) (6%). Opening up multiple worker channels also avoided the restriction imposed by Mechanical Turk in 2013 to limit the labor pool to workers based in the US and India. Full details along with the range of trust scores for coders from these platforms are presented in the supplementary materials (section 4).

[26] Full point estimates are provided in the supplementary materials, section 1.

[27] An alternative measure of correlation, Lin's concordance correlation coefficient (Lin 1989, 2000), measures correspondence as well covariation, if our objective is to match the values on the identity line, although for many reasons here it is not. The economic and social measures for Lin's coefficient are 0.95 and 0.84, respectively.

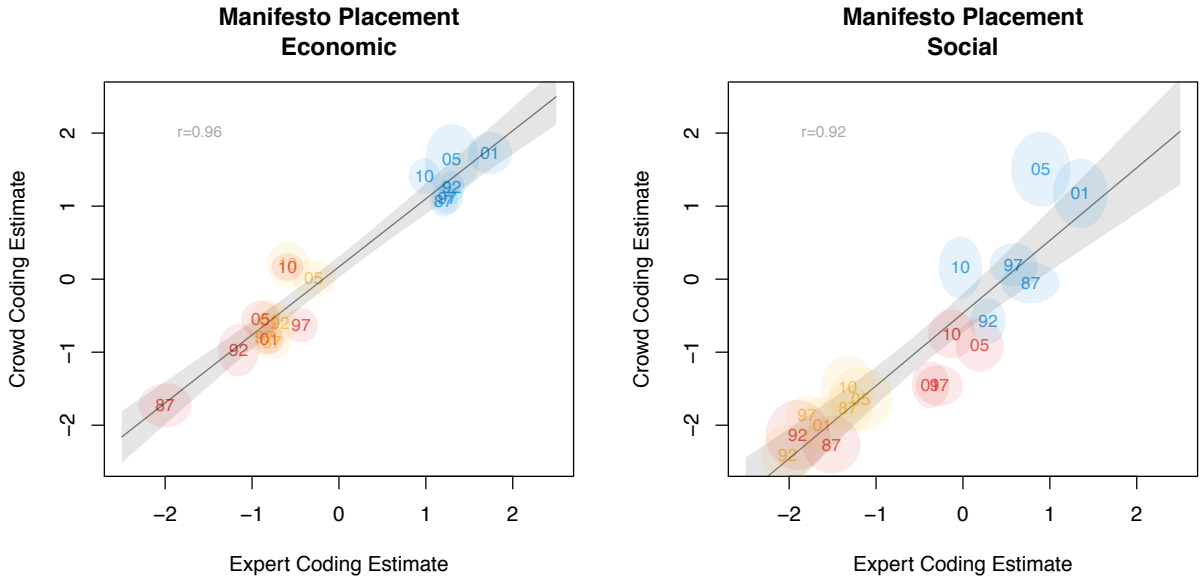


Figure 3. Expert and crowd-sourced estimates of economic and social policy positions.

Our scaling model provides a theoretically well-grounded way to aggregate all the information in our expert or crowd data, relating the underlying position of the political text both to the "difficulty" of a particular sentence and to a reader's propensity to identify the correct policy domain, and position within domain.[28] Because positions derived from the scaling model depend on parameters estimated using the full set of coders and codings, changes to the text corpus can affect the relative scaling. The simple mean of means method, however, is invariant to rescaling and always produces the same results, even for a single document. Comparing crowd-sourced estimates from the scaling model to those produced by a simple averaging of the mean of mean sentence scores, we find correlations of 0.96 for the economic and 0.97 for the social policy positions of the 18 manifestos. We present both methods as confirmation that our scaling method has not "manufactured" policy estimates. While this model does allow us to take proper account of reader and sentence fixed effects, it is also reassuring that a simple mean of means produced substantively similar estimates.

[28] We report more fully on diagnostic results for our coders on the basis of the auxiliary model quantity estimates in the supplementary materials (section 1e).

We have already seen that noisy expert judgments about sentences aggregate up to reliable and valid estimates for documents. Similarly, crowd-sourced document estimates reported in Figure 3 are derived from crowd-sourced sentence data that are full of noise. As we already argued, this is the essence of crowd-sourcing. Figure 4 plots mean expert and against mean crowd-sourced scores *for each sentence*. The scores are highly correlated, though crowd workers are substantially less likely to use extremes of the scales than experts. The first principal component and associated confidence intervals show a strong and significant statistical relationship between crowd sourced and expert assessments of individual manifesto sentences, with no evidence of systematic bias in the crowd-coded sentence scores.[29] Overall, despite the expected noise, our results show that crowd workers systematically tend to make the same judgments about individual sentences as experts.

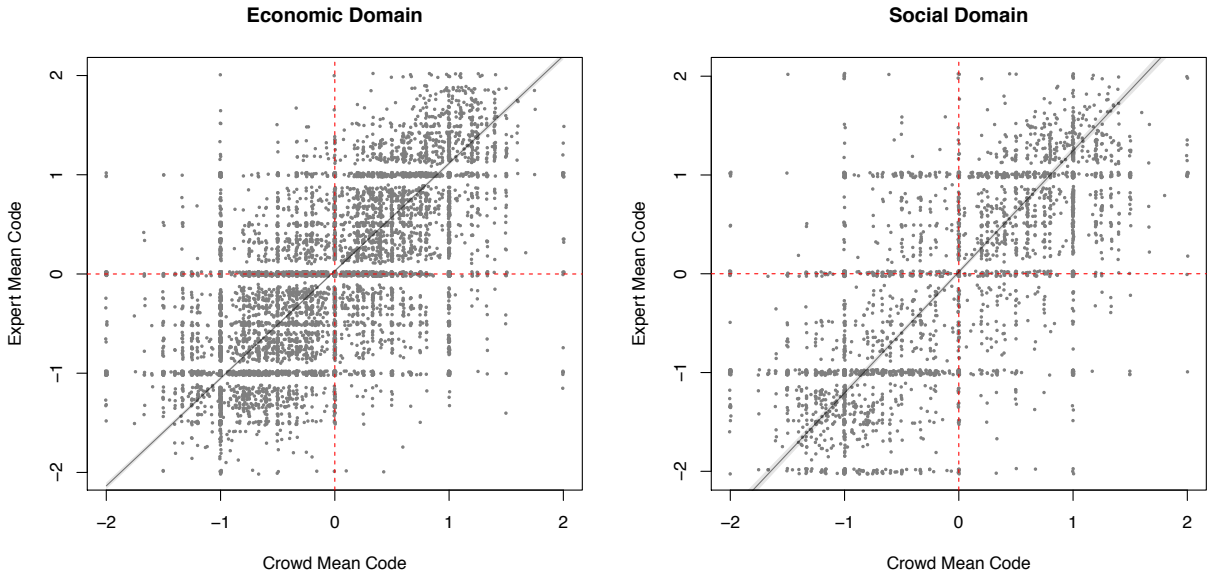


Figure 4. Expert and crowd-sourced estimates of economic and social policy codes of individual sentences, all manifestos. Fitted line is the principal components or Deming regression line.

[29] Lack of bias is indicated by the fact that the fitted line crosses the origin.

Calibrating the number of crowd judgments per sentence

A key question for our method concerns *how many* noisier crowd-based judgments we need to generate reliable and valid estimates of fairly long documents such as party manifestos. To answer this, we turn to evidence from our over-sampling of 1987 and 1997 manifestos. Recall we obtained a minimum of 20 crowd judgments for each sentence in each of these manifestos, allowing us to explore what our estimates of the position of each manifesto would have been, had we collected fewer judgments. Drawing random subsamples from our over-sampled data, we can simulate the convergence of estimated document positions as a function of the number of crowd judgments per sentence. We did this by bootstrapping 100 sets of subsamples for each of the subsets of n=1 to n=20 workers, computing manifesto positions in each policy domain from aggregated sentence position means, and computing standard deviations of these manifesto positions across the 100 estimates. Figure 5 plots these for each manifesto as a function of the increasing number of crowd workers per sentence, where each point represents the empirical standard error of the estimates for a specific manifesto. For comparison, we plot the same quantities for the expert data in red.

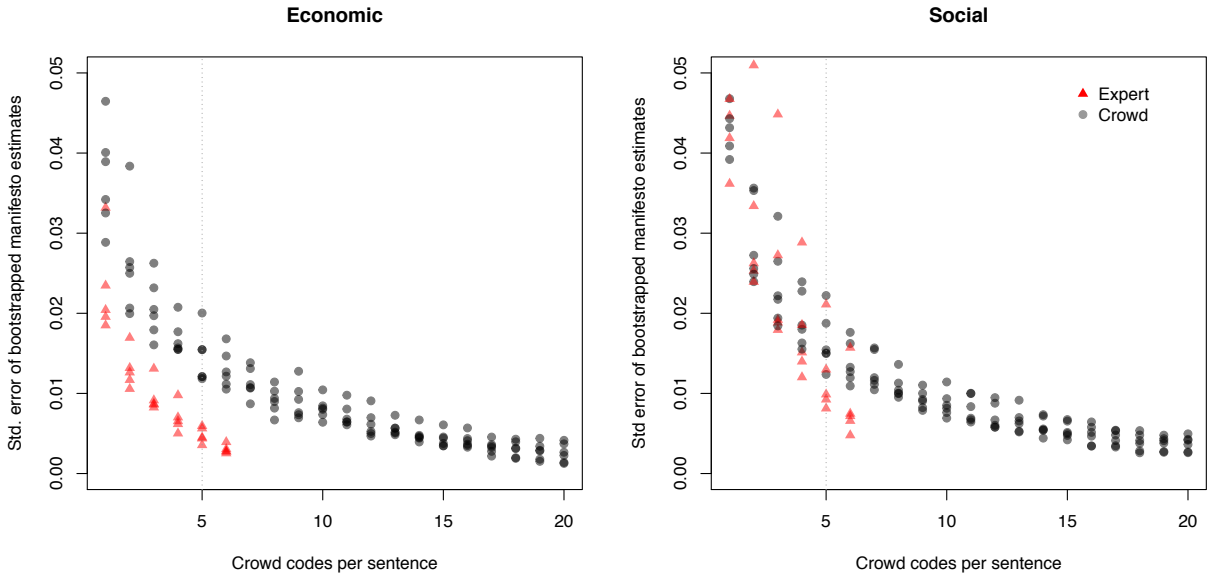


Figure 5. Standard errors of manifesto-level policy estimates as a function of the number of workers, for the oversampled 1987 and 1997 manifestos. Each point is the bootstrapped standard deviation of the mean of means aggregate manifesto scores, computed from sentence-level random *n* sub-samples from the codes.

The findings show a clear trend: uncertainty over the crowd-based estimates collapses as we increase the number of workers per sentence. Indeed, the only difference between experts and the crowd is that expert variance is smaller, as we would expect. Our findings vary somewhat with policy area, given the noisier character of social policy estimates, but adding additional crowd-sourced sentence judgments led to convergence with our expert panel of 5-6 coders at around 15 crowd coders. However, the steep decline in the uncertainty of our document estimates leveled out at around five crowd judgments per sentence, at which point the absolute level of error is already low for both policy domains. While increasing the number of unbiased crowd judgments will always give better estimates, we decided on cost-benefit grounds for the second stage of our deployment to continue coding in the crowd until we had obtained five crowd judgments per sentence. This may seem a surprisingly small number, but there are a number of important factors to bear in mind in this context. First, the manifestos comprise about

1000 sentences on average; our estimates of document positions aggregate codes for these. Second, sentences were randomly assigned to workers, so each sentence score can be seen as an independent estimate of the position of the manifesto on each dimension.[30] With five scores per sentence and about 1000 sentences per manifesto, we have about 5000 "little" estimates of the manifesto position, each a representative sample from the larger set of scores that would result from additional worker judgments about each sentence in each document. This sample is big enough to achieve a reasonable level of precision, given the large number of sentences per manifesto. While the *method* we use here could be used for much shorter documents, the *results* we infer here for the appropriate number of judgments per sentence might well not apply, and would likely be higher. But, for large documents with many sentences, we find that the number of crowd judgments *per sentence* that we need is not high.

CROWD-SOURCING DATA FOR SPECIFIC PROJECTS: IMMIGRATION POLICY

A key problem for scholars using "canonical" datasets, over and above the replication issues we discuss above, is that the data often do not measure what a modern researcher wants to measure. For example the widely-used MP data, using a classification scheme designed in the 1980s, do not measure immigration policy, a core concern in the party politics of the 21st century (Ruedin and Morales 2012; Ruedin 2013). Crowd-sourcing data frees researchers from such "legacy" problems and allows them more flexibly to collect information on their precise quantities of interest. To demonstrate this, we designed a project tailored to measure British parties' immigration policies during the 2010 election. We analyzed the manifestos of eight parties, including smaller parties with more extreme positions on immigration, such as the British National Party (BNP) and the UK Independence Party (UKIP). Workers were asked to label each

[30] Coding a sentence as referring to another dimension is a null estimate.

sentence as referring to immigration policy or not. If a sentence did cover immigration, they were asked to rate it as pro- or anti-immigration, or neutral. We deployed a job with 7,070 manifesto sentences plus 136 "gold" questions and screeners devised specifically for this purpose. For this job, we used an adaptive sentence sampling strategy which set a minimum of five crowd sourced labels per sentence, unless the first three of these were unanimous in judging a sentence *not* to concern immigration policy. This is efficient when coding texts with only "sparse" references to the matter of interest; in this case most manifesto sentences (approximately 96%) were clearly not about immigration policy. Within just five hours, the job was completed, with 22,228 codings, for a total cost of $360.[31]

We assess the external validity of our results using independent expert surveys by Benoit (2010) and the Chapel Hill Expert Survey (Marks 2010). Figure 6 compares the crowd-sourced estimates to those from expert surveys. The correlation with the Benoit (2010) estimates (shown) was 0.96, and 0.94 with independent expert survey estimates from the Chapel Hill survey.[32] To assess whether this data production exercise was as reproducible as we claim, we repeated the entire exercise with a second deployment two months after the first, with identical settings. This new job generated another 24,551 pieces of crowd-sourced data and completed in just over three hours. The replication generated nearly identical estimates, detailed in Table 4, correlating at the same high levels with external expert surveys, and correlating at 0.93 with party position estimates from the original crowd-coding.[33] With just hours from deployment to dataset, and for very little cost, crowd-sourcing enabled us to generate externally valid *and reproducible* data related to our precise research question.

[31] The job set 10 sentences per "task" and paid $0.15 per task.
[32] CHES included two highly correlated measures, one aimed at "closed or open" immigration policy another aimed at policy toward asylum seekers and whether immigrants should be integrated into British society. Our measure averages the two. Full numerical results are given in supplementary materials, section 3.
[33] Full details are in the supplementary materials, section 7.

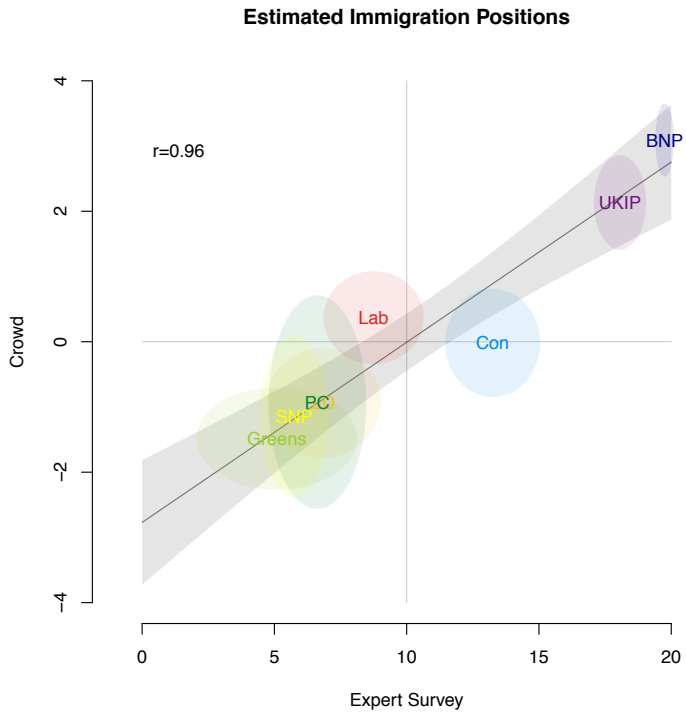


Figure 6. Correlation of combined immigration crowd codings with Benoit (2010) expert survey position on immigration.

	Wave		
	Initial	Replication	Combined
Total Crowd Codings	24,674	24,551	49,225
Number of Coders	51	48	85
Total sentences coded as Immigration	280	264	283
Correlation with Benoit expert survey (2010)	0.96	0.94	0.96
Correlation with CHES 2010	0.94	0.91	0.94
Correlation of results between waves			0.93

Table 4. Comparison results for Replication of Immigration Policy Crowd-Coding.

CROWD SOURCED TEXT ANALYSIS IN OTHER CONTEXTS AND LANGUAGES

As carefully designed official statements of a party's policy stances, election manifestos tend to respond well to systematic text analysis. In addition, manifestos are written for popular consumption and tend to be easily understood by non-technical readers. Much political information, however, can be found in texts generated from hearings, committee debates, or legislative speeches on issues that often refer to technical provisions, amendments, or other rules of procedure that might prove harder to analyze. Furthermore, a majority of the world's political texts are not in English. Other widely studied political contexts, such as the European Union, are multi-lingual environments where researchers using automated methods designed for a single language must make hard choices. Schwarz et al. (Forthcoming) applied unsupervised scaling methods to a multilingual debate in the Swiss parliament, for instance, but had to ignore a substantial number of French and Italian speeches in order to focus on the majority German texts. In this section, we demonstrate that crowd-sourced text analysis, with appropriately translated instructions, offers the means to overcome these limitations by working in any language.

Our corpus comes from a debate in the European Parliament, a multi-language setting where the EU officially translates every document into 22 languages. To test our method in a context very different from party manifestos, we chose a fairly technical debate concerning a Commission report proposing an extension to a regulation permitting state aid to uncompetitive coal mines. This debate concerned not only the specific proposal, involving a choice of letting the subsidies expire in 2011, permitting a limited continuation until 2014, or extending them

until 2018 or even indefinitely.[34] It also served as debating platform for arguments supporting state aid to uncompetitive industries, versus the traditionally liberal preference for the free market over subsidies. Because a vote was taken at the end of the debate, we also have an objective measure of whether the speakers supported or objected to the continuation of state aid.

We downloaded all 36 speeches from this debate, originally delivered by speakers from 11 different countries in 10 different languages. Only one of these speakers, an MEP from the Netherlands, spoke in English, but all speeches were officially translated into each target language. After segmenting this debate into sentences, devising instructions and representative test sentences, we deployed the same text analysis job in English, German, Spanish, Italian, Polish, and Greek, using crowd workers to read and label the same set of texts, but using the translation into their own language. Figure 7 plots the score for each text against the eventual vote of the speaker. It shows that our crowd-sourced scores for each speech perfectly predict the voting behavior of each speaker, regardless of the language. In Table 5, we show correlations between our crowd-sourced estimates of the positions of the six different language versions of the same set of texts. The results are striking, with inter-language correlations ranging between 0.92 and 0.96.[35] Our text measures from this technical debate produced reliable measures of the very specific dimension we sought to estimate, and the validity of these measures was demonstrated by their ability to predict the voting behavior of the speakers. Not only are these results straightforwardly reproducible, but this reproducibility is invariant to the language in which the speech was written. Crowd-sourced text analysis does not only work in English.

[34] This was the debate from 23 November 2010, "State aid to facilitate the closure of uncompetitive coal mines." http://www.europarl.europa.eu/sides/getDoc.do?pubRef=-//EP//TEXT+CRE+20101123+ITEM-005+DOC+XML+V0//EN&language=EN

[35] Lin's concordance coefficient has a similar range of values, from 0.90 to 0.95.

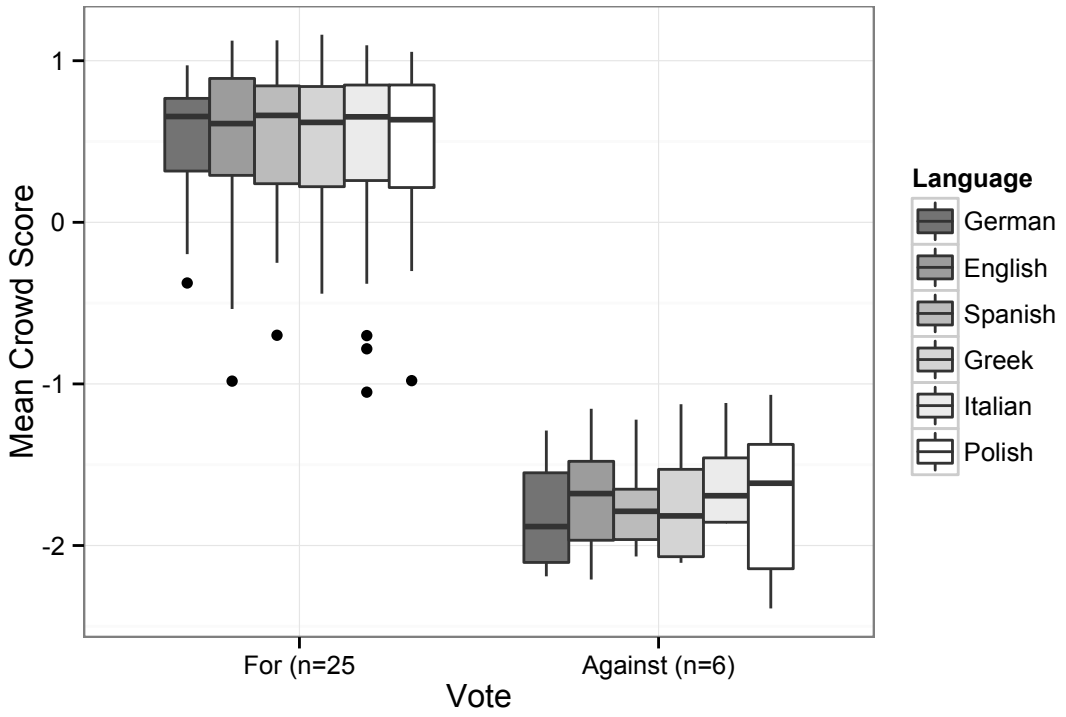


Figure 7. Scored speeches from a debate over state subsidies by vote, from separate crowd-sourced text analysis in six languages. Aggregate scores are standardized for direct comparison.

	Correlations of 35 speaker scores					
Language	English	German	Spanish	Italian	Greek	Polish
German	0.96	--	--	--	--	--
Spanish	0.94	0.95	--	--	--	--
Italian	0.92	0.94	0.92	--	--	--
Greek	0.95	0.97	0.95	0.92	--	--
Polish	0.96	0.95	0.94	0.94	0.93	--
Sentence *N*	414	455	418	349	454	437
Total Judgments	3,545	1,855	2,240	1,748	2,396	2,256
Cost	$109.33	$55.26	$54.26	$43.69	$68.03	$59.25
Elapsed Time (hrs)	1	3	3	7	2	1

Table 5. Summary of Results from EP Debate Coding in 6 languages

CONCLUSIONS

We demonstrated across a range of applications that crowd-sourced text analysis can produce valid political data of a quality indistinguishable from traditional expert methods. Unlike traditional methods, however, crowd-sourced data generation offers several key advantages. Foremost among these is the possibility of meeting a standard of replication far stronger than the current practice of facilitating reproducible *analysis*. By offering a published specification for replicating *the process of data generation*, the methods demonstrated here go much farther in meeting a more stringent standard of *reproducibility* that is the hallmark of scientific inquiry. All of the data used in this paper are of course available in a public archive for any reader to reanalyze at will. Crowd-sourcing our data allows us to do much more than this, however. Any reader can take our publically available crowdsourcing code and deploy this code to *reproduce our data collection process and collect a completely new dataset*. This can be done many times over, by any researcher, anywhere in the world, with only the limited resources needed to pay the workers in the crowd. This, to our minds, takes us significantly closer to a true scientific replication standard.

Another key advantage of crowd-sourced text analysis is that the process of development and deployment can form part of an *agile* research process, tailored to a research question rather than representing a grand compromise designed to meet as many present and future needs as can be anticipated, given the enormous fixed costs of traditional data collection that tends to discourage repetition or adaptation. Because the crowd's resources can be tapped in a flexible fashion, text-based data can be processed only for the contexts, questions, and time periods required. Coupled with the rapid completion time of crowd-sourced tasks and their low cost, this opens the possibility of valid text processing to researchers with limited resources, especially

graduate students. For those with more ambition or resources, its scalability means that crowd-sourcing can tackle large projects as well. In our demonstrations, it worked as well for hundreds of judgments as it did for hundreds of thousands.

One very promising application of crowd-sourcing for political science is to integrate automated methods for text analysis with human judgment, a relatively new field known as "active learning" (Arora and Agarwal 2007). Crowd-sourcing can both make automated approaches more effective, as well as solve problems that quantitative text analysis cannot (yet) address. Supervised methods require labeled training data and unsupervised methods require *a posteriori* interpretation, both of which can be provided by either experts or by the crowd. But for many tasks that require human interpretation, particularly of information embedded in the syntax of language rather than in the bag of words that are used, untrained human coders in the crowd provide a combination of human intelligence and affordability that neither computers nor experts can beat.

APPENDIX: METHODOLOGICAL DECISIONS ON SERVING POLITICAL TEXT TO WORKERS IN THE CROWD

900 words

Text units: natural sentences

The MP specifies a "quasi-sentence" as the fundamental text unit, defined as "an argument which is the verbal expression of one political idea or issue" (Volkens). Recoding experiments by Däubler et al. (2012), however, show that using natural sentences makes no statistically significant difference to point estimates, but does eliminate significant sources of both unreliability and unnecessary work. Our dataset therefore consists of all natural sentences in the 18 UK party manifestos under investigation.[36]

Text unit sequence: random

In "classical" expert text coding, experts process sentences in their natural sequence, starting at the beginning and ending at the end of a document. Most workers in the crowd, however, will never reach the end of a long policy document. Processing sentences in natural sequence, moreover, creates a situation in which one sentence coding may well affect priors for subsequent sentence codings, so that summary scores for particular documents are not aggregations of independent coder assessments.[37] An alternative is to randomly sample sentences from the text corpus for coding—with a fixed number of replacements per sentence across all coders—so that each coding is an independent estimate of the latent variable of interest. This has the big advantage in a crowdsourcing context of *scalability*. Jobs for individual coders can range from very small to very large; coders can pick up and put down coding tasks at will; every little piece

[36] Segmenting "natural" sentences, even in English, is never an exact science, but our rules matched those from Däubler et al. (2012), treating (for example) separate clauses of bullet pointed lists as separate sentences.
[37] Coded sentences do indeed tend to occur in "runs" of similar topics, and hence codes; however to ensure appropriate statistical aggregation it is preferable if the codings of those sentences are independent.

of coding in the crowd contributes to the overall database of text codings. Accordingly our method for crowd-sourced text coding serves coders sentences randomly selected from the text corpus rather than in naturally occurring sequence. Our decision to do this was informed by coding experiments reported in the supplementary materials (section 5), and confirmed by results reported below. Despite higher variance in individual sentence codings under random sequence coding, there is no systematic difference between point estimates of party policy positions depending on whether sentences were coded in natural or random sequence.

Text authorship: anonymous

In classical expert coding, coders typically know the authorship of the document they are coding. Especially in the production of political data, coders likely bring non-zero priors to coding text units. Precisely the same sentence ("we must do all we can to make the public sector more efficient") may be coded in different ways if the coder knows this comes from a right- rather than a left-wing party. Codings are typically aggregated into document scores as if coders had zero priors, even though we do not know how much of the score given to some sentence is the coder's judgment about the content of the sentence, and how much a judgment about its author. In coding experiments reported in supplementary materials (section 5), semi-expert coders coded the same manifesto sentences both knowing and not knowing the name of the author. We found slight systematic coding biases arising from knowing the identity of the document's author. For example, we found coders tended to code precisely the same sentences from Conservative manifestos as more right wing, if they knew these sentences came from a Conservative manifesto. This informed our decision to withhold the name of the author of sentences deployed in crowd-sourcing text coding.

Context units: +/- two sentences

Classical content analysis has always involved coding an individual text unit in light of the text surrounding it. Often, it is this context that gives a sentence substantive meaning, for example because many sentences contain pronoun references to surrounding text. For these reasons, careful instructions for drawing on context have long formed part of coder instructions for content analysis (see Krippendorff 2013). For our coding scheme, on the basis of pre-release coding experiments, we situated each "target" sentence within a context of the two sentences either side in the text. Coders were instructed to code target sentence not context, but to use context to resolve any ambiguity they might feel about the target sentence.

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CROWD-SOURCED TEXT ANALYSIS:

SUPPLEMENTAL MATERIALS

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These supplementary results contain additional information on the crowd-coding, the expert coding, the semi-expert testing, and our scaling diagnostics for the economic and social results. They also contain full instructions and codes required to replicate our jobs on Crowdflower. Additional materials required such as the sentence datasets and original texts are available in our replication materials, including Stata and R code required to transform the texts into data for Crowdflower, and to analyze judgments from Crowdflower.

CONTENTS

1. Economic and Social Scaling Estimates 1

2. JAGS code for model estimation 6

3. Expert survey estimates 9

4. Details on the Crowd Coders 11

5. Details on pre-testing the deployment method using semi-expert coders 14

6. Implementation and Instructions for Econ/Social Jobs on CrowdFlower 19

7. Instructions for "Coding sentences from a parliamentary debate" 27

8. Full crowd-sourced coding results for immigration policy 30

9. Instructions for the crowd-sourced analysis of the first round reviews 31

1. Economic and Social Scaling Estimates

a) Expert v. crowd

	Expert		Crowd	
Manifesto	Economic	Social	Economic	Social
Con 1987	1.21	0.78	1.07	-0.05
	[0.9, 1.23]	[-0.27, 0.17]	[0.9, 1.23]	[-0.27, 0.17]
LD 1987	-0.77	-1.32	-0.87	-1.77
	[-1.03, -0.68]	[-2.03, -1.52]	[-1.03, -0.68]	[-2.03, -1.52]
Lab 1987	-2.00	-1.51	-1.73	-2.27
	[-1.99, -1.51]	[-2.6, -1.98]	[-1.99, -1.51]	[-2.6, -1.98]
Con 1992	1.25	0.58	1.26	-0.58
	[0.94, 1.28]	[-0.04, 0.43]	[1.09, 1.46]	[-0.83, -0.34]
LD 1992	-0.85	-1.79	-0.59	-2.41
	[-0.99, -0.63]	[-2.07, -1.64]	[-0.83, -0.39]	[-2.71, -2.06]
Lab 1992	-0.43	-0.27	-0.97	-2.13
	[-0.82, -0.46]	[-1.68, -1.24]	[-1.24, -0.66]	[-2.46, -1.75]
Con 1997	1.30	0.29	1.11	0.20
	[1.09, 1.46]	[-0.83, -0.34]	[0.94, 1.28]	[-0.04, 0.43]
LD 1997	-0.68	-2.01	-0.79	-1.87
	[-0.83, -0.39]	[-2.71, -2.06]	[-0.99, -0.63]	[-2.07, -1.64]
Lab 1997	-1.15	-1.90	-0.63	-1.46
	[-1.24, -0.66]	[-2.46, -1.75]	[-0.82, -0.46]	[-1.68, -1.24]
Con 2001	1.74	1.35	1.73	1.18
	[1.49, 1.95]	[0.78, 1.55]	[1.49, 1.95]	[0.78, 1.55]
LD 2001	-0.80	-1.62	-0.55	-2.00
	[-0.76, -0.33]	[-2.28, -1.68]	[-0.76, -0.33]	[-2.28, -1.68]
Lab 2001	-0.80	-0.37	-0.82	-1.44
	[-0.98, -0.67]	[-1.73, -1.19]	[-0.98, -0.67]	[-1.73, -1.19]
Con 2005	1.30	0.90	1.64	1.51
	[1.29, 2.03]	[1.08, 1.9]	[1.29, 2.03]	[1.08, 1.9]
LD 2005	-0.29	-1.18	0.01	-1.64
	[-0.2, 0.22]	[-1.98, -1.31]	[-0.2, 0.22]	[-1.98, -1.31]
Lab 2005	-0.90	0.19	-0.55	-0.90
	[-0.73, -0.35]	[-1.2, -0.64]	[-0.73, -0.35]	[-1.2, -0.64]
Con 2010	0.98	-0.03	1.41	0.16
	[1.18, 1.59]	[-0.14, 0.52]	[1.18, 1.59]	[-0.14, 0.52]
LD 2010	-0.59	-1.33	0.20	-1.48
	[-0.05, 0.46]	[-1.8, -1.15]	[-0.05, 0.46]	[-1.8, -1.15]
Lab 2010	-0.59	-0.13	0.17	-0.75
	[0.02, 0.32]	[-1.04, -0.5]	[0.02, 0.32]	[-1.04, -0.5]
Correlation with Expert Survey Estimates	0.91	0.82	0.91	0.74
Correlation with Expert Mean of Means	1.00	0.99	1.00	1.00

Table 1. Model Estimates for Expert-coded Positions on Economic and Social Policy.

b) Comparing expert sequential versus random order sentence coding

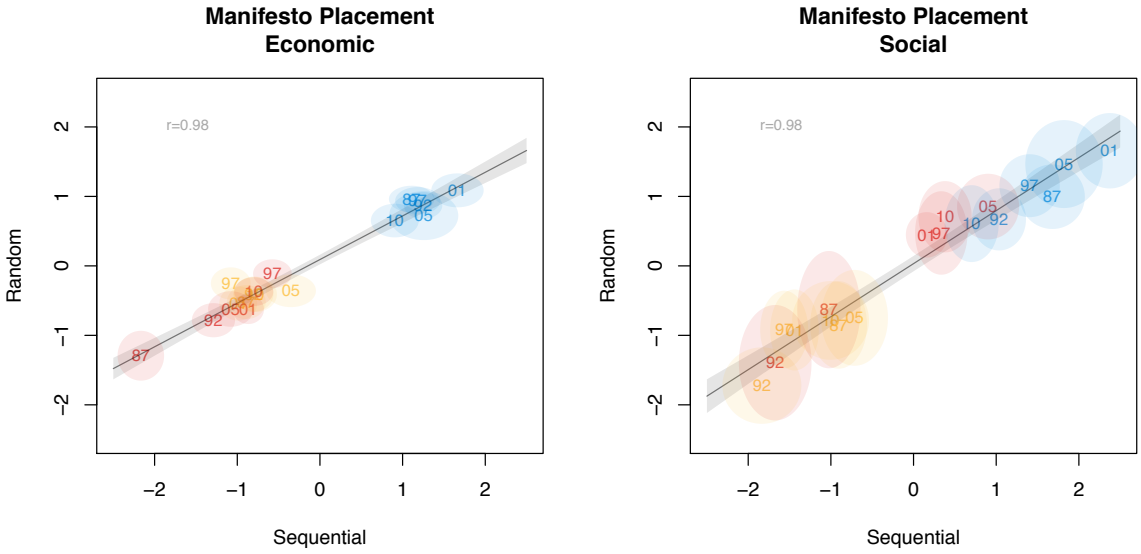


Figure 1. Scale estimates from expert coding, comparing expert sequential and unordered sentence codings.

c) Cronbach's alpha for social policy scale

Item	*N*	*Sign*	*Item-scale correlation*	*Item-rest correlation*	*Cronbach's alpha*
Expert 1	959	+	0.93	0.79	0.92
Expert 2	431	+	0.87	0.78	0.95
Expert 3	579	+	0.92	0.78	0.95
Expert 4	347	+	0.91	0.79	0.94
Expert 5	412	+	0.90	0.78	0.94
Expert 6	211	+	0.92	0.86	0.93
Overall					0.95

Table 2. Inter-coder reliability analysis for the social policy scale generated by aggregating all expert scores for sentences judged to have social policy content.

This table provides the social policy scale equivalent for Table 3 of the main paper.

d) Coder-level diagnostics from economic and social policy coding

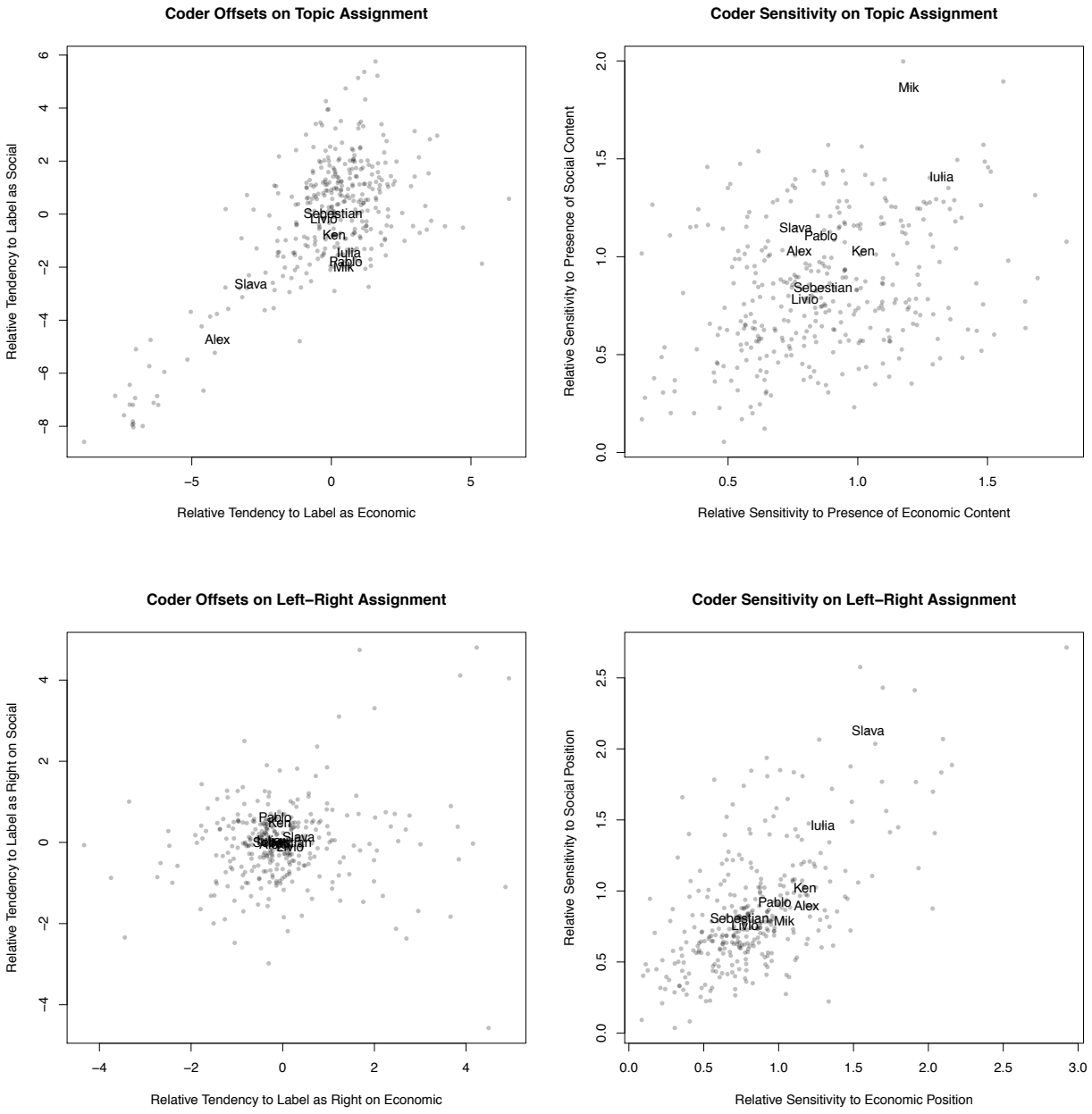


Figure 2. Coder-level parameters for expert coders (names) and crowd coders (points). Top plots show offsets ψ_{id} and sensitivities χ_{id} in assignment to social and economic categories versus none; bottom plots show offsets and sensitivities in assignment to left-right scale positions.

e) Convergence diagnostics

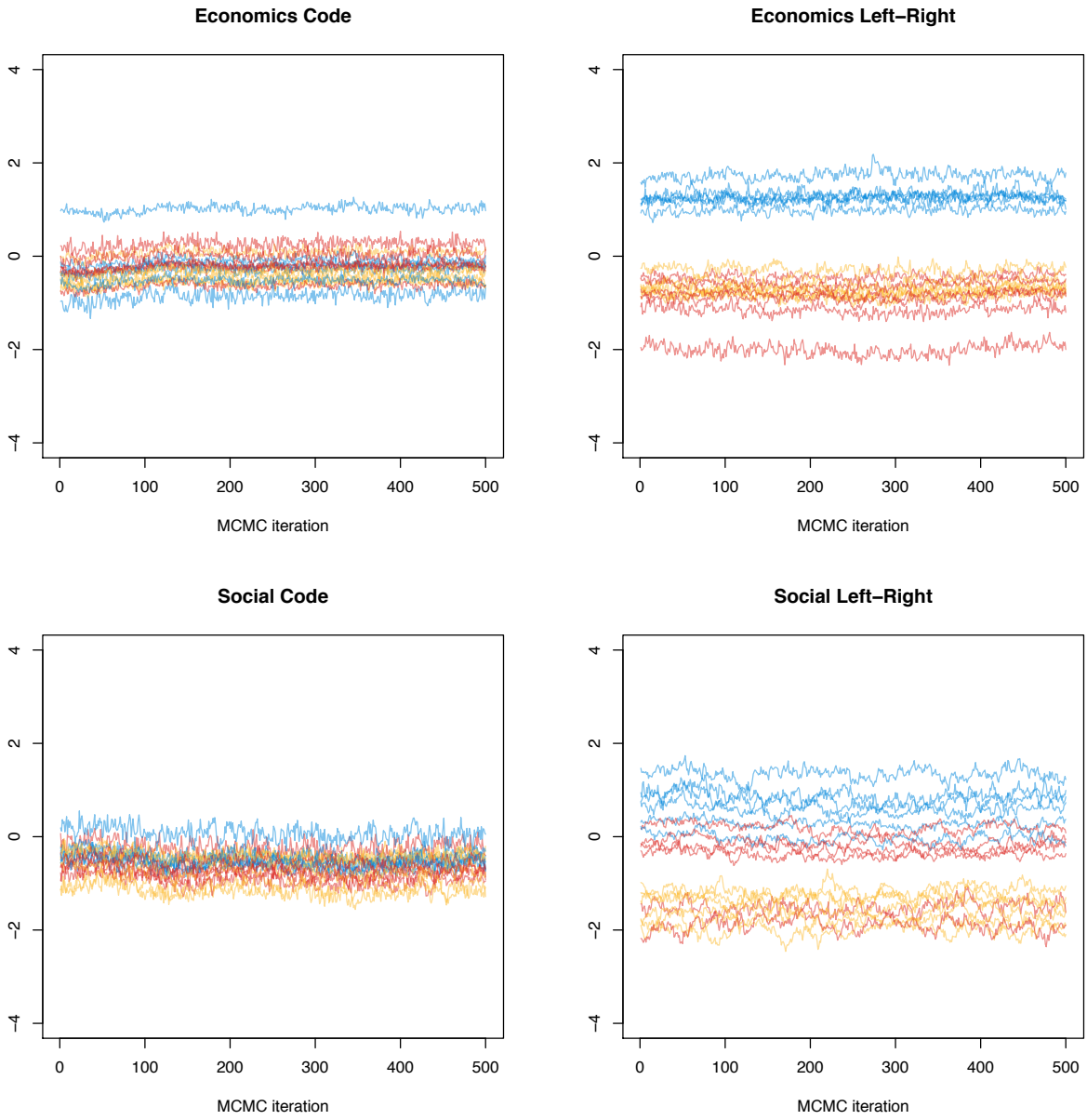


Figure 3. MCMC trace plots for manifesto-level parameters for expert coders.

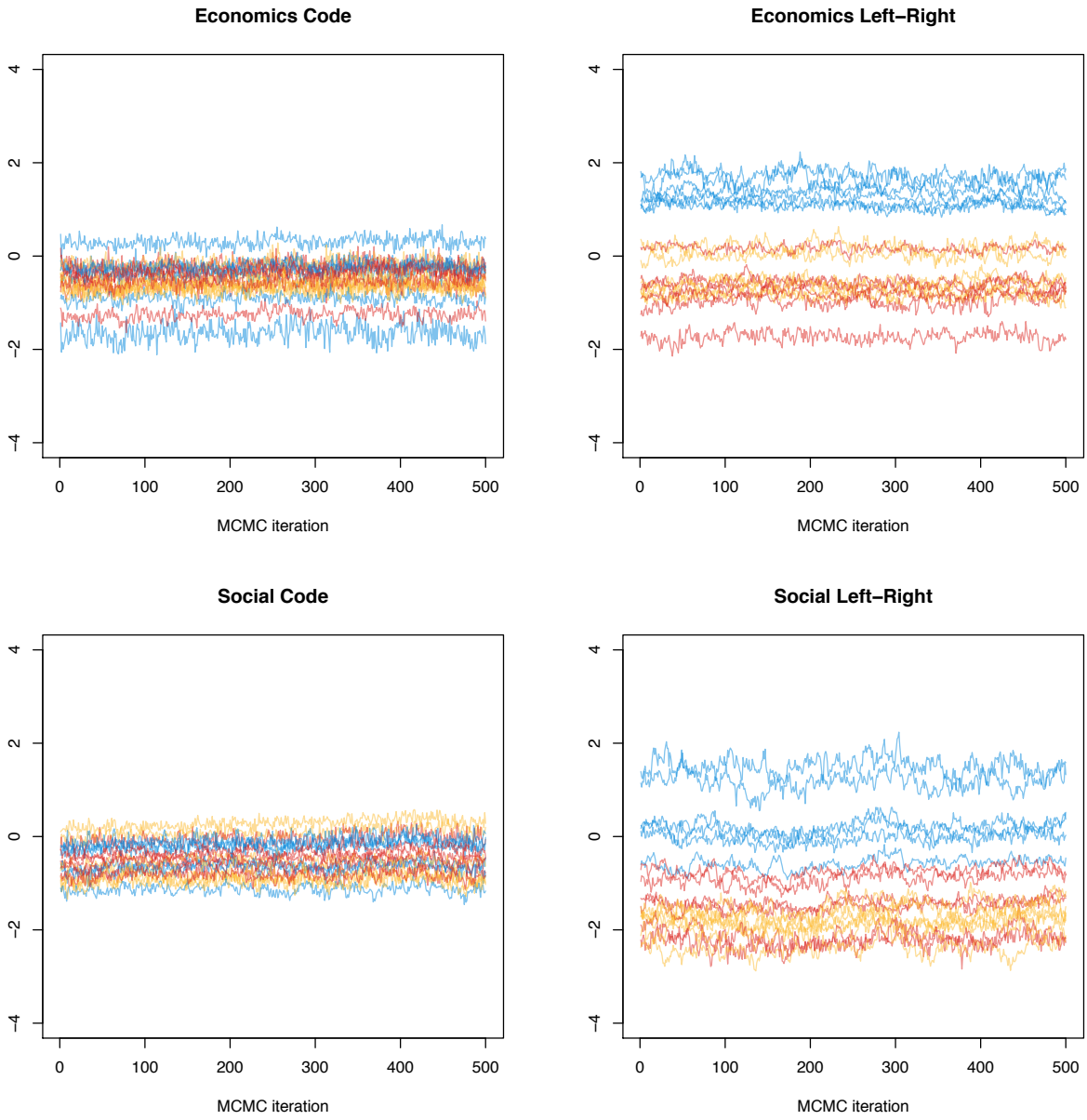


Figure 4. MCMC trace plots for manifesto-level parameters for crowd coders.

2. JAGS code for model estimation

a) Economic and social policy scaling

```
model {

    for (q in 1:Ncodings){

        # Define latent response for code/scale in econ/social

        mucode[q,1] <- (theta[SentenceID[q],1,1] + psi[CoderID[q],1,1])*chi[CoderID[q],1,1];
        mucode[q,2] <- (theta[SentenceID[q],2,1] + psi[CoderID[q],2,1])*chi[CoderID[q],2,1];
        muscale[q,1] <- (theta[SentenceID[q],1,2] + psi[CoderID[q],1,2])*chi[CoderID[q],1,2];
        muscale[q,2] <- (theta[SentenceID[q],2,2] + psi[CoderID[q],2,2])*chi[CoderID[q],2,2];

        # Translate latent responses into 11 category probabilities (up to normalization)

        mu[q,1] <- 1;
        mu[q,2] <- exp(mucode[q,1])*(ilogit(-1*cut[2] - muscale[q,1]));
        mu[q,3] <- exp(mucode[q,1])*(ilogit(-1*cut[1] - muscale[q,1])-ilogit(-1*cut[2] - muscale[q,1]));
        mu[q,4] <- exp(mucode[q,1])*(ilogit(1*cut[1] - muscale[q,1])-ilogit(-1*cut[1] - muscale[q,1]));
        mu[q,5] <- exp(mucode[q,1])*(ilogit(1*cut[2] - muscale[q,1])-ilogit(1*cut[1] - muscale[q,1]));
        mu[q,6] <- exp(mucode[q,1])*(1-ilogit(1*cut[2] - muscale[q,1]));
        mu[q,7] <- exp(mucode[q,2])*(ilogit(-1*cut[2] - muscale[q,2]));
        mu[q,8] <- exp(mucode[q,2])*(ilogit(-1*cut[1] - muscale[q,2])-ilogit(-1*cut[2] - muscale[q,2]));
        mu[q,9] <- exp(mucode[q,2])*(ilogit(1*cut[1] - muscale[q,2])-ilogit(-1*cut[1] - muscale[q,2]));
        mu[q,10] <- exp(mucode[q,2])*(ilogit(1*cut[2] - muscale[q,2])-ilogit(1*cut[1] - muscale[q,2]));
        mu[q,11] <- exp(mucode[q,2])*(1-ilogit(1*cut[2] - muscale[q,2]));

        # 11 category multinomial

        Y[q] ~ dcat(mu[q,1:11]);

    }

    # Specify uniform priors for ordinal thresholds (assumes left-right symmetry)

    cut[1] ~ dunif(0,5);
    cut[2] ~ dunif(cut[1],10);

    # Priors for coder bias parameters

    for (i in 1:Ncoders) {
        psi[i,1,1] ~ dnorm(0,taupsi[1,1]);
        psi[i,2,1] ~ dnorm(0,taupsi[2,1]);
        psi[i,1,2] ~ dnorm(0,taupsi[1,2]);
        psi[i,2,2] ~ dnorm(0,taupsi[2,2]);
    }

    # Priors for coder sensitivity parameters

    for (i in 1:Ncoders) {
        chi[i,1,1] ~ dnorm(0,1)T(0,);
        chi[i,2,1] ~ dnorm(0,1)T(0,);
        chi[i,1,2] ~ dnorm(0,1)T(0,);
```

```
		chi[i,2,2] ~ dnorm(0,1)T(0,);
	}

	# Priors for sentence latent parameters

	for (j in 1:Nsentences) {
		theta[j,1,1] ~ dnorm(thetabar[ManifestoIDforSentence[j],1,1],tautheta[1,1]);
		theta[j,2,1] ~ dnorm(thetabar[ManifestoIDforSentence[j],2,1],tautheta[2,1]);
		theta[j,1,2] ~ dnorm(thetabar[ManifestoIDforSentence[j],1,2],tautheta[1,2]);
		theta[j,2,2] ~ dnorm(thetabar[ManifestoIDforSentence[j],2,2],tautheta[2,2]);
	}

	# Priors for manifesto latent parameters

	for (k in 1:Nmanifestos) {
		thetabar[k,1,1] ~ dnorm(0,1);
		thetabar[k,2,1] ~ dnorm(0,1);
		thetabar[k,1,2] ~ dnorm(0,1);
		thetabar[k,2,2] ~ dnorm(0,1);
	}

	# Variance parameters

	taupsi[1,1] ~ dgamma(1,1);
	taupsi[2,1] ~ dgamma(1,1);
	taupsi[1,2] ~ dgamma(1,1);
	taupsi[2,2] ~ dgamma(1,1);

	tautheta[1,1] ~ dgamma(1,1);
	tautheta[2,1] ~ dgamma(1,1);
	tautheta[1,2] ~ dgamma(1,1);
	tautheta[2,2] ~ dgamma(1,1);

}
```

b) Immigration policy scaling

```
model {

	for (q in 1:Ncodings){

		# Define latent response for code/scale in econ/social

		mucode[q] <- (theta[SentenceID[q],1] + psi[CoderID[q],1])*chi[CoderID[q],1];
		muscale[q] <- (theta[SentenceID[q],2] + psi[CoderID[q],2])*chi[CoderID[q],2];

		# Translate latent responses into 4 category probabilities (up to normalization)

		mu[q,1] <- 1;
		mu[q,2] <- exp(mucode[q])*(ilogit(-1*cut[1] - muscale[q]));
		mu[q,3] <- exp(mucode[q])*(ilogit(1*cut[1] - muscale[q])-ilogit(-1*cut[1] - muscale[q]));
		mu[q,4] <- exp(mucode[q])*(1-ilogit(1*cut[1] - muscale[q]));

		# 11 category multinomial
```

```
            Y[q] ~ dcat(mu[q,1:4]);

        }

        # Specify uniform priors for ordinal thresholds (assumes left-right symmetry)

        cut[1] ~ dunif(0,10);

        # Priors for coder bias parameters

        for (i in 1:Ncoders) {
            psi[i,1] ~ dnorm(0,taupsi[1]);
            psi[i,2] ~ dnorm(0,taupsi[2]);
        }

        # Priors for coder sensitivity parameters

        for (i in 1:Ncoders) {
            chi[i,1] ~ dnorm(0,1)T(0,);
            chi[i,2] ~ dnorm(0,1)T(0,);
        }

        # Priors for sentence latent parameters

        for (j in 1:Nsentences) {
            theta[j,1] ~ dnorm(thetabar[ManifestoIDforSentence[j],1],tautheta[1]);
            theta[j,2] ~ dnorm(thetabar[ManifestoIDforSentence[j],2],tautheta[2]);
        }

        # Priors for manifesto latent parameters

        for (k in 1:Nmanifestos) {
            thetabar[k,1] ~ dnorm(0,1);
            thetabar[k,2] ~ dnorm(0,1);
        }

        # Variance parameters

        taupsi[1] ~ dgamma(1,1);
        taupsi[2] ~ dgamma(1,1);

        tautheta[1] ~ dgamma(1,1);
        tautheta[2] ~ dgamma(1,1);

}
```

3. Expert survey estimates

These are taken from Laver and Hunt (1992); Laver (1998) for 1997; Benoit and Laver (2006) for 2001; Benoit (2005, 2010) for 2005 and 2010. For reference and because the results from Benoit (2005, 2010) were never published, we produce them here.

Party	Party Name	Year	Dimension	Mean	N	SE
Con	Conservative Party	1987	Economic	17.2	34	0.40
Lab	Labour Party	1987	Economic	5.4	34	0.38
LD	Liberal Democrats	1987	Economic	8.2	34	0.43
PCy	Plaid Cymru	1987	Economic	5.4	34	0.48
SNP	Scottish National Party	1987	Economic	6.0	34	0.42
Con	Conservative Party	1997	Economic	15.1	117	0.23
Lab	Labour Party	1997	Economic	10.3	117	0.23
LD	Liberal Democrats	1997	Economic	5.8	116	0.23
PCy	Plaid Cymru	1997	Economic	5.2	89	0.25
SNP	Scottish National Party	1997	Economic	5.6	92	0.26
Con	Conservative Party	2001	Economic	15.3	56	0.40
Lab	Labour Party	2001	Economic	8.1	57	0.40
LD	Liberal Democrats	2001	Economic	5.8	57	0.37
PCy	Plaid Cymru	2001	Economic	5.2	45	0.39
SNP	Scottish National Party	2001	Economic	6.1	46	0.49
BNP	British National Party	2005	Economic	10.5	26	0.73
Con	Conservative Party	2005	Economic	14.9	76	0.27
Lab	Labour Party	2005	Economic	8.1	77	0.27
LD	Liberal Democrats	2005	Economic	5.3	77	0.28
PCy	Plaid Cymru	2005	Economic	5.2	57	0.30
SNP	Scottish National Party	2005	Economic	5.8	59	0.33
UKIP	UK Independence Party	2005	Economic	14.8	26	0.67
BNP	British National Party	2010	Economic	9.6	16	1.09
Con	Conservative Party	2010	Economic	15.8	19	0.46
GPEW	Green Party of England and Wales	2010	Economic	4.9	17	0.58
Lab	Labour Party	2010	Economic	6.8	19	0.43
LD	Liberal Democrats	2010	Economic	11.5	20	0.79
PCy	Plaid Cymru	2010	Economic	4.5	15	0.45
SNP	Scottish National Party	2010	Economic	5.9	16	0.77
SSP	Scottish Socialist Party	2010	Economic	2.2	15	0.37
UKIP	UK Independence Party	2010	Economic	14.3	18	1.10

Table 3. Expert Survey Estimates of UK Political Parties, Economic Dimension.

Party	Party Name	Year	Dimension	Mean	N	SE
Con	Conservative Party	1987	Social	15.3	34	0.45
Lab	Labour Party	1987	Social	6.5	34	0.36
LD	Liberal Democrats	1987	Social	6.9	34	0.41
PCy	Plaid Cymru	1987	Social	9.0	34	0.44
SNP	Scottish National Party	1987	Social	9.6	34	0.31
Con	Conservative Party	1997	Social	13.3	116	0.25
Lab	Labour Party	1997	Social	8.3	116	0.23
LD	Liberal Democrats	1997	Social	6.8	113	0.24
PCy	Plaid Cymru	1997	Social	9.4	84	0.27
SNP	Scottish National Party	1997	Social	9.4	87	0.25
Con	Conservative Party	2001	Social	15.3	57	0.33
Lab	Labour Party	2001	Social	6.9	57	0.32
LD	Liberal Democrats	2001	Social	4.1	56	0.24
PCy	Plaid Cymru	2001	Social	7.7	37	0.48
SNP	Scottish National Party	2001	Social	8.1	37	0.41
BNP	British National Party	2005	Social	18.0	35	0.31
Con	Conservative Party	2005	Social	14.5	77	0.28
Lab	Labour Party	2005	Social	7.3	77	0.30
LD	Liberal Democrats	2005	Social	4.4	77	0.20
PCy	Plaid Cymru	2005	Social	7.4	44	0.42
SNP	Scottish National Party	2005	Social	7.8	44	0.37
UKIP	UK Independence Party	2005	Social	16.3	24	0.52
BNP	British National Party	2010	Social	18.3	17	0.39
Con	Conservative Party	2010	Social	12.4	18	0.69
GPEW	Green Party of England and Wales	2010	Social	3.1	16	0.44
Lab	Labour Party	2010	Social	5.0	18	0.62
LD	Liberal Democrats	2010	Social	3.8	18	0.42
PCy	Plaid Cymru	2010	Social	7.1	14	0.82
SNP	Scottish National Party	2010	Social	6.5	14	0.71
SSP	Scottish Socialist Party	2010	Social	3.3	9	0.50
UKIP	UK Independence Party	2010	Social	14.7	15	0.81
BNP	British National Party	2010	Immigration	19.8	16	0.14
Con	Conservative Party	2010	Immigration	13.3	16	0.73
GPEW	Green Party of England and Wales	2010	Immigration	5.1	11	1.24
Lab	Labour Party	2010	Immigration	8.8	16	0.77
LD	Liberal Democrats	2010	Immigration	6.9	15	0.88
PCy	Plaid Cymru	2010	Immigration	6.6	8	0.75
SNP	Scottish National Party	2010	Immigration	5.8	8	0.53
SSP	Scottish Socialist Party	2010	Immigration	4.6	5	0.40
UKIP	UK Independence Party	2010	Immigration	18.1	16	0.40
UKIP	UK Independence Party	2010	Economic	14.3	18	1.10

Table 4. Expert Survey Estimates of UK Political Parties, Social and Immigration Dimensions.

4. Details on the Crowd Coders

Country	Total Codings	% Codings	Unique Coders	Mean Trust Score
USA	60,117	28.0	697	0.85
GBR	33,031	15.4	199	0.84
IND	22,739	10.6	91	0.79
ESP	12,284	5.7	4	0.76
EST	10,685	5.0	4	0.87
DEU	9,934	4.6	43	0.86
HUN	9,402	4.4	36	0.83
HKG	7,955	3.7	29	0.89
CAN	7,028	3.3	112	0.85
POL	6,425	3.0	47	0.83
HRV	4,487	2.1	45	0.79
A1	3,612	1.7	1	0.83
AUS	2,667	1.2	15	0.80
MEX	2,607	1.2	41	0.80
ROU	2,565	1.2	7	0.84
NLD	2,466	1.2	14	0.82
PAK	1,908	0.9	4	0.80
IDN	1,860	0.9	8	0.76
CZE	1,804	0.8	16	0.81
GRC	1,803	0.8	1	0.71
SRB	1,718	0.8	2	0.79
LTU	1,163	0.5	16	0.83
DOM	779	0.4	1	0.83
ZAF	722	0.3	8	0.82
ITA	666	0.3	5	0.81
IRL	628	0.3	7	0.83
MKD	606	0.3	2	0.77
ARG	534	0.3	2	0.90
BGR	513	0.2	2	0.90
DNK	497	0.2	9	0.83
VNM	413	0.2	1	0.82
TUR	400	0.2	2	0.75
PHL	268	0.1	4	0.79
FIN	253	0.1	8	0.88
PRT	139	0.1	3	0.86
MAR	86	0.0	2	1.00
MYS	79	0.0	3	0.85
Other (12)	264	0.0	12	0.83
Overall	215,107	100.0	1503	0.83

Table 5. Country Origins and Trust Scores of the Crowd Coders.

Channel	Total Codings	% Codings	Trust Score	
			Mean	95% CI
Neodev	85,991	39.98	0.83	[0.83, 0.83]
Amt	39,288	18.26	0.85	[0.84, 0.85]
Bitcoinget	32,124	14.93	0.88	[0.88, 0.88]
Clixsense	28,063	13.05	0.81	[0.81, 0.81]
Prodege	12,151	5.65	0.83	[0.83, 0.83]
Probux	5,676	2.64	0.83	[0.83, 0.83]
Instagc	4,166	1.94	0.81	[0.81, 0.81]
Rewardingways	2,354	1.09	0.89	[0.89, 0.90]
Coinworker	1,611	0.75	0.90	[0.90, 0.90]
Taskhunter	1,492	0.69	0.81	[0.80, 0.81]
Taskspay	881	0.41	0.78	[0.78, 0.78]
Surveymad	413	0.19	0.82	[0.82, 0.82]
Fusioncash	303	0.14	0.81	[0.80, 0.82]
Getpaid	253	0.12	0.81	[0.81, 0.82]
Other (12)	341	0.16	0.89	[0.88, 0.91]
Total	215,107	100.00	0.84	[0.84, 0.84]

Table 6. Crowdflower Crowd Channels and Associated Mean Trust Scores.

Job ID	Date Launched	Sentences	Gold Sentences	Codings		Sentences Per Task	Payment Per Task	Minimum Codings Per Sentence	Mean Cost Per Trusted Code	Cost	Dimension	Countries	Channels
				Trusted	Untrusted								
389381	18-Feb-14	7,206	136	22118	1287	10	$0.12	3	$0.02	$390.60	Immigration	Many	Many
354277	10-Dec-13	7,206	136	22228	681	10	$0.15	3	$0.02	$537.07	Immigration	Many	Many
354285	11-Dec-13	12,819	551	65101	10442	8	$0.12	5	$0.03	$1,676.73	Econ/Social	Many	Many
313595	10-Nov-13	129	18	258	12	8	$0.12	2	$0.03	$7.38	Econ/Social	Many	Many
303590	31-Oct-13	194	27	970	107	8	$0.12	5	$0.03	$27.72	Econ/Social	Many	Many
302426	30-Oct-13	2,314	330	11577	3214	8	$0.12	5	$0.03	$331.56	Econ/Social	Many	Many
302314	30-Oct-13	239	33	2394	285	5	$0.08	10	$0.03	$79.59	Econ/Social	Many	Many
299749	29-Oct-13	1,506	165	22638	8364	10	$0.20	15	$0.04	$814.90	Econ/Social	Many	Many
269506	23-Oct-13	2,400	300	24239	7064	10	$0.20	10	$0.04	$877.80	Econ/Social	Many	Many
263548	22-Oct-13	901	99	13519	2787	10	$0.20	15	$0.04	$487.67	Econ/Social	Many	Many
246609	01-Oct-13	55	30	550	940	10	$0.20	10	$0.05	$27.64	Econ/Social	US	MT
246554	01-Oct-13	901	85	9016	3951	10	$0.20	10	$0.04	$320.56	Econ/Social	US	MT
240807	23-Sep-13	452	48	2264	207	10	$0.20	5	$0.04	$81.28	Econ/Social	US	MT
139288	17-Oct-12	684	43	3487	106	5	$0.11	5	$0.04	$146.25	Econ/Social	US	MT
131743	27-Sep-12	10	43	80	91	5	$0.28	5	$0.34	$27.14	Econ/Social	All	MT
131742	27-Sep-12	87	43	484	290	5	$0.28	5	$0.14	$66.57	Econ/Social	All	MT
131741	27-Sep-12	297	43	1614	2790	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131740	27-Sep-12	297	43	1596	2470	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131739	27-Sep-12	297	43	1574	2172	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131738	27-Sep-12	297	43	1581	2110	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131737	27-Sep-12	297	43	1613	2140	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131736	27-Sep-12	297	43	1594	2135	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131735	27-Sep-12	297	43	1616	987	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131733	27-Sep-12	297	43	1567	1064	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131732	27-Sep-12	297	43	1576	1553	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131731	27-Sep-12	297	43	1585	1766	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
131562	26-Sep-12	297	43	1534	1054	5	$0.28	5	$0.11	$174.10	Econ/Social	All	MT
130980	24-Sep-12	297	43	1618	1041	5	$0.15	5	$0.06	$93.27	Econ/Social	All	MT
130148	20-Sep-12	259	504	1436	372	5	$0.06	5	$0.06	$83.73	Econ/Social	All	MT
130147	20-Sep-12	259	504	672	798	5	$0.06	5	$0.12	$83.73	Econ/Social	All	MT

Table 7. Details on Phased Crowdflower Job Deployments for Economic and Social Text Policy Coding.

5. Details on pre-testing the deployment method using semi-expert coders

Design

Our design of the coding platform followed several key requirements of crowd-sourcing, namely that the coding be split into sentence-level tasks with clear instructions, aimed only at the specific policy dimensions we have already identified. This involved several key decisions, which we settled on following extensive tests on expert coders (including the authors and several PhD-level coders with expertise in party politics) and semi-experts consisting of trained postgraduate students given a set of experimental texts where the features being tested were varied in an experimental context to generate results to detect the design with the highest reliability and validity. These decisions were: whether to serve the sentences in sequence or randomly; whether to identify the document being coded; and how many contextual sentences to display for the sentence.

Sequential versus unordered sentences

In what we call "classical" expert coding, experts typically start at the beginning of a document and work through, sentence by sentence, to the end.[1] From a practical point of view, however, most workers in the crowd will code only small sections of an entire long policy document. From a theoretical point of view, moreover, coding sentences in their natural sequence creates a situation in which coding one sentence may well affect priors for subsequent sentence codings, with the result that some sentence codings may be affected by how immediately preceding sentences have been coded. In particular, sentences in sequence tend to display "runs" of similar topics, and hence codes, given the natural tendency of authors to organize a text into clusters of similar topics. To mitigate the tendency of coders to also pass judgment on each text unit in runs without considering each sentence on the grounds of its own content, we tested whether text coding produced more stable results when served up unordered rather than in the sequence of the text.

Anonymous texts versus named texts

In serving up sentence coding tasks, another option is whether to identify the texts by name, or instead for them to remain anonymous.2 Especially in relation to a party manifesto, it is not necessary to read very far into the document, even if cover and title page have been ripped off, to figure out which party wrote it – indeed we might reasonably deem a coder who cannot figure this out to be unqualified. Coders will likely bring non-zero priors to coding manifesto sentences: precisely the same sentence ("we must do all we can to make the public sector more efficient") may be coded in different ways if the coder knows this comes from a right- rather than a left-wing party. Yet codings are typically aggregated into estimated document scores as if coders had zero priors. We don't really know how much of the score given to any given sentence in classical expert coding is the coder's judgment about the actual content of the sentence, and how much is

[1] We may leave open the sequence in which *documents* are coded, or make explicit decisions about this, such as coding according to date of authorship.

[2] Of course, many of the party manifestos we used made references to their own party names, making it fairly obvious which party wrote the manifesto. In these cases we did not make any effort to anonymize the text, as we did with to risk altering the meaning.

a judgment about its author. Accordingly, in our preliminary coding experiment, expert coders coded the same manifesto sentences both knowing and not knowing the name of the author.

Providing context for the target sentence

Given the results we note in the previous two sections, our crowd-sourcing method will specify the atomic crowd-sourced text coding task as coding a "target" sentence selected at random from a text, with the name of the author not revealed. This leaves open the issue of how much context either side of the target sentence we provide to assist the coder. The final objective of our preliminary coding experiment was to assess the effects of providing no context at all, or a one- or two- sentence context either side of the target. To test the effects on reliability, our pre-test experiments provided the same sentences, in random order, to the semi-expert coders with zero, one, and two sentences of context before and after the sentence to be coded.

Results of the pre-testing

We pre-tested the coding scheme decisions on a sample of three co-authors of this paper, three additional expert coders trained personally, by the authors, and 30 "semi-expert" coders who were Masters students in courses on applied text analysis at either XXX or XXX. (The detailed design for the administration of treatments to coder is available from the authors.) To assess coder reliability, we also created a carefully agreed set of 120 "gold standard" sentences whose codes were unanimously agreed by the expert coders. Using an experimental design in which each coder in the test panel coded each sentence multiple times, in random order, with variation across the three treatment effects, we gathered sufficient information to predict misclassification tendencies from the coding set using a multinomial logistic model. The results pointed to a minimization of misclassification by: a) serving up codings tasks with unordered sentences, b) not identifying the author of the text, and c) providing two sentences of context before and after each sentence to be coded. The most significant finding was that coders had a mild but significant tendency to code the same sentences differently when they associated the known author of the text with a particular position. Specifically, they tended to code precisely the same sentences from Conservative manifestos as more right wing, if they knew that these sentences came from a Conservative manifesto. We also found a slight but significantly better correspondence between coder judgments and "golden" codings when we provided a context of two sentences before and after the sentence to be coded. This informed out decision to settle on a two-sentence context for our crowd-sourcing method.

The aim of this methodological experiment was to assess effects of: coding manifestos in their natural sequence or in random order (Treatment 1); providing a +/- two-sentence context for the target sentence (Treatment 2); revealing the title of the manifesto and hence the name of its author (Treatment 3). The text corpus to be coded was a limited but carefully-curated set of 120 sentences. We removed some surrounding sentences that had proper party names in them, to maintain a degree of manifesto anonymity. These were chosen on the basis of the classical expert coding (ES) phase of our work to include a balance of sentences between expert-coded economic and social policy content, and only a few sentences with no economic or social policy content. The coder pool comprised three expert coders, three co-authors of this paper, and 30 "semi-expert" coders who were Masters students in Methods courses at either XXX or XXX. The detailed design for the administration of treatments to coder is available from the authors. The analysis depends in part on the extent to which the "semi-expert" coders agreed with a master or

"gold" coding for each sentence, which we specified as the majority scale and code from the three "expert" coders.

For each sentence that was master-coded as referring to none, economic, or social policy, Table 8 reports exponentiated coefficients from a multinomial logit predicting how a coder would classify a sentence, using the sentence variables as covariates. This allows direct computation of misclassification, given a set of controls. Since all variables are binary, we report odds ratios. Thus the highlighted coefficient of 3.272 in Model 1 means that, when the master coding says the sentence concerns neither economic nor social policy, the odds of a coder misclassifying the sentence as economic policy were about 3.3 times higher if the sentence displayed a title, all other things held constant. More generally, we see from Table 8 that providing a +/- two-sentence context does tend to reduce misclassifications (with odds ratios less that 1.0) while showing the coder the manifesto title does tend to increase misclassification (with odds ratios greater than 1.0).

Confining the data to sentence codings for which the coder agreed with the master coding on the policy area covered by the sentence, Table 9 reports an ordinal logit of the positional codes assigned by non-expert coders, controlling for fixed effects of the manifesto. The base category is the relatively centrist Liberal Democrat manifesto of 1987. The main quantities of interest estimate the interactions of the assigned positional codes with title and context treatments. If there is no effect of title or context, then these interactions should add nothing. If revealing the title of the manifesto makes a difference, this should for example move economic policy codings to the left for a party like Labour, and to the right for the Conservatives. The highlighted coefficients show that this is a significant effect, though only for Conservative manifestos.

		(1)	(2)	(3)
			Master Domain	
Equation	Independent Variable	Neither	Economic	Social
Economic	Context	0.492*		2.672
		(0.214 - 1.132)		(0.702 - 10.18)
	Sequential	1.069		0.896
		(0.578 - 1.978)		(0.396 - 2.030)
	Title	3.272***		1.053
		(2.010 - 5.328)		(0.532 - 2.085)
Social	Context	0.957	0.822	
		(0.495 - 1.850)	(0.583 - 1.160)	
	Sequential	0.867	1.05	
		(0.527 - 1.428)	(0.800 - 1.378)	
	Title	1.540**	1.064	
		(1.047 - 2.263)	(0.877 - 1.291)	
None	Context		0.478***	0.643
			(0.280 - 0.818)	(0.246 - 1.681)
	Sequential		1.214	2.598**
			(0.758 - 1.943)	(1.170 - 5.766)
	Title		0.854	0.807
			(0.629 - 1.159)	(0.505 - 1.292)
	N	750	3,060	1,590

Odds ratios (95% confidence intervals), *** p<0.01, ** p<0.05, * p<0.1

Table 8. Domain Misclassification in Semi-Expert Coding Experiments.

	(4)	(5)	(6)	(7)
	Coded [-1, 0, 1]		Coded [-2, -1, 0, 1, 2]	
Independent Variable	Economic	Social	Economic	Social
Con 1987	8.541***	158.7***	9.939***	286.8***
	(4.146 - 17.60)	(79.86 - 315.4)	(4.050 - 24.39)	(87.86 - 936.4)
Lab 1987	0.867	0.902	1.066	2.268
	(0.386 - 1.946)	(0.409 - 1.993)	(0.444 - 2.556)	(0.478 - 10.77)
Con 1997	5.047***	4.248***	4.385***	10.80***
	(2.485 - 10.25)	(1.754 - 10.29)	(2.063 - 9.320)	(2.919 - 39.97)
LD 1997	0.953		1.089	
	(0.493 - 1.841)		(0.546 - 2.171)	
Lab 1997	3.274***	328.0***	4.554***	1,004***
	(1.623 - 6.604)	(146.1 - 736.5)	(2.087 - 9.941)	(246.1 - 4,099)
Context	0.386***	1.113	0.389***	1.218
	(0.218 - 0.685)	(0.719 - 1.724)	(0.211 - 0.719)	(0.408 - 3.637)
Context * Con 1987	2.675**	0.834	3.425**	0.972
	(1.225 - 5.841)	(0.414 - 1.682)	(1.258 - 9.327)	(0.270 - 3.497)
Context * Lab 1987	0.62	2.772**	0.373**	3.184
	(0.263 - 1.463)	(1.114 - 6.895)	(0.144 - 0.968)	(0.592 - 17.12)
Context * Con 1997	3.734***	1.106	3.713***	0.805
	(1.806 - 7.719)	(0.422 - 2.900)	(1.716 - 8.036)	(0.193 - 3.362)
Context * LD 1997	2.785***		2.645***	
	(1.395 - 5.557)		(1.280 - 5.468)	
Context * Lab 1997	1.008	0.855	0.846	0.713
	(0.487 - 2.088)	(0.425 - 1.721)	(0.378 - 1.894)	(0.184 - 2.763)
Title	0.506***	0.857	0.557**	0.87
	(0.331 - 0.773)	(0.585 - 1.256)	(0.346 - 0.896)	(0.326 - 2.320)
Title * Con 1987	1.920**	1.133	2.309**	1.252
	(1.114 - 3.306)	(0.614 - 2.089)	(1.105 - 4.825)	(0.393 - 3.983)
Title * Lab 1987	1.211	0.672	1.16	0.954
	(0.639 - 2.295)	(0.350 - 1.293)	(0.510 - 2.639)	(0.299 - 3.041)
Title * Con 1997	1.891**	2.080*	1.446	2.492
	(1.086 - 3.292)	(0.971 - 4.457)	(0.778 - 2.690)	(0.734 - 8.459)
Title * LD 1997	1.35		1.205	
	(0.793 - 2.299)		(0.675 - 2.149)	
Title * Lab 1997	1.439	0.618	1.236	0.549
	(0.826 - 2.505)	(0.347 - 1.101)	(0.676 - 2.260)	(0.169 - 1.787)
Sequential	0.842	0.84	0.843	0.802
	(0.680 - 1.044)	(0.639 - 1.104)	(0.658 - 1.080)	(0.529 - 1.218)
Observations	2,370	1,481	2,370	1,481

Table 9. Scale bias in semi-expert coding experiments.

6. Implementation and Instructions for Econ/Social Jobs on CrowdFlower

Once gold data have been identified, CF has a flexible system for working with many different types of crowd-sourcing task. In our case, preparing the manifesto texts for CF coders requires converting the text into a matrix-organized dataset with one natural sentence per row. CF uses its own proprietary markup language, CrowdFlower Markup Language (CML), to build jobs on the platform. The language is based entirely on HTML, and contains only a small set of special features that are needed to link the data being used for the job to the interface itself. To create the coding tasks themselves, some additional markup is needed. Here we use two primary components: a text chunk to be coded, and the coding interface. To provide context for the text chunk, we include two sentences of preceding and proceeding manifesto text, in-line with the sentence being coded. The line to be coded is colored red to highlight it. The data are then linked to the job using CML, and the CF platform will then serve up the coding tasks as they appear in the dataset. To design the interface itself we use CML to design the form menus and buttons, but must also link the form itself to the appropriate data. Unlike the sentence chunk, however, for the interface we need to tell the form which columns in our data will be used to store the workers' coding; rather than where to pull data from. In addition, we need to alert the CF platform as to which components in the interface are used in gold questions.

Coding economic and social policy statements in political text (20131210)

Instructions ▲

This task involves reading sentences from political texts and judging whether these deal with economic or social policy.

The sentences you will be asked about come from political party manifestos. Some deal with economic policy, some with social policy; others deal with neither economic nor social policy. We tell you below what we mean by "economic" and "social" policy.

First, you will read a short section from a party manifesto. For the sentence highlighted in red, enter your best judgment about whether it mainly refers to economic policy, to social policy, or to neither.

If the sentence refers to economic policy, select "economic" in the drop down menu; if it refers to social policy, select "social". If the sentence does not refer to either policy area, select "not economic or social" – in this case you will move directly to the next sentence.

If you select "economic" or "social" policy, you will be asked to give your best judgment of the sentence how much it is left or right wing (for economic policy), or liberal or conservative (for social policy). We tell you below about what we mean by "left" and "right" etc.

For example, if you see a sentence with what you think is very right wing economic policy, select the economic policy area from the drop down menu and click "very right" on the economic policy scale. If you think it has a position that is left-wing, but not very left wing, click "left".

If you believe the sentence expresses a centrist position on economic or social policy OR concerns economic or social policy but does not express any clear position, select the appropriate policy category from the drop down menu, as above, and click the "neither…nor…" position on the scale.

Now we will tell you about what we mean by "economic" and "social" policy, and by "left" and "right", "liberal" and "conservative".

What is "economic" policy? What are "left" or "right" economic policies?

"Economic" policies deal with all aspects of the economy, including:

- Taxation and government spending
- Services provided by government or other public bodies
- Pensions, unemployment and welfare benefits; other state benefits
- Property, investment and share ownership, public or private
- Interest rates and exchange rates
- Regulation of economic activity, public or private
- Relations between employers, workers and trade unions

"Left" economic policies tend to favor one or more of the following:

- High levels of state benefits and services provided by the government, even if these imply high levels of taxation;
- Public investment; public ownership or control of sections of business and industry;
- Public regulation of private business and economic activity;
- Support for workers/trade unions relative to employers

"Right" economic policies tend to favor one or more of the following:

- Low levels of taxation, even if this implies low levels of levels of state benefits and services provided by the government;
- Private investment; minimal public ownership or control of business and industry;
- Minimal public regulation of private business and economic activity;
- Support for employers relative to trade unions/workers

Examples

Below we provide two examples of text from the manifestos and instructions on how they should be coded, and why.

Example 1: "Right" economic policy:

With a Conservative Government, all that has been changing. **We were determined to make share-ownership available to the whole nation.** Just as with cars, television sets, washing machines and foreign holidays, it would no longer be a privilege of the few

The highlighted text should be coded as "economic" policy because it refers to ownership; and as on the "right" because it promotes private ownership.

Example 2: "Liberal" social policy:

ALTERNATIVES TO PRISON. Every effort should be made to ensure that fine defaulters, elderly shoplifters and drunks are not sent to prison. **Police cautions and intermediate treatment should be more widely used.** Where punishment is appropriate, it should normally be community service rather than prison

The highlighted text should be coded as having to do with "social" policy because refers to policing, and as "liberal" because it promotes alternatives to prison.

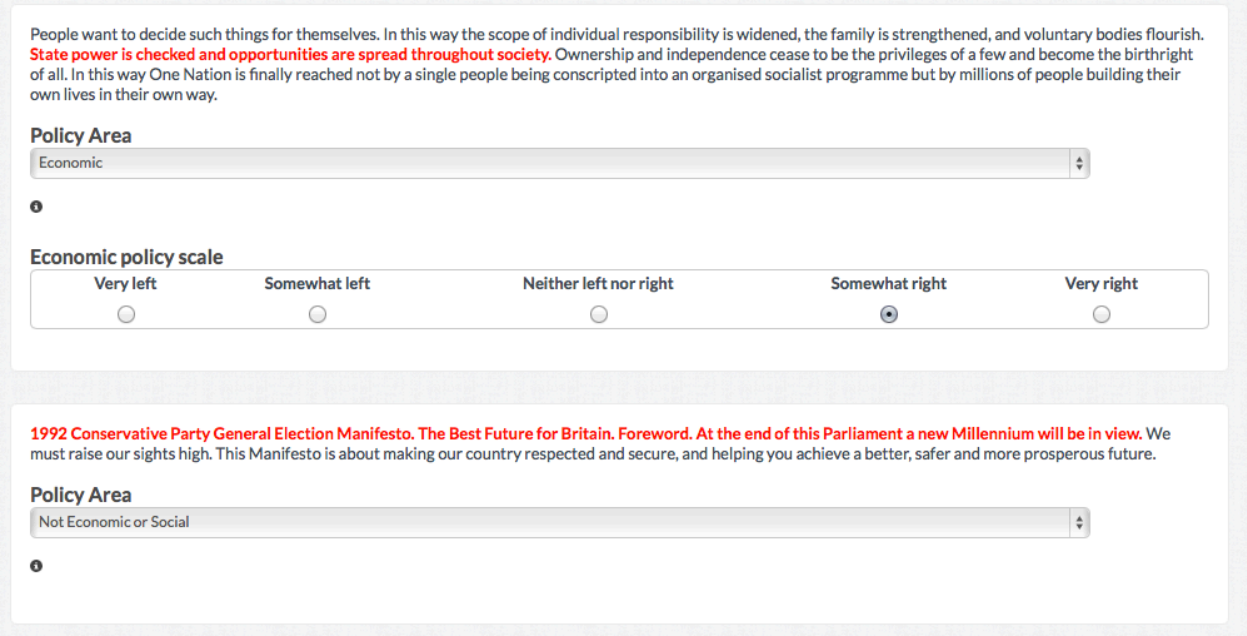


Figure 5a. Screenshot of text coding platform, implemented in CrowdFlower.

The above images show a screen shot of the coding interface as deployed and Figure A2 shows the CML used to design our this interface. With all aspects of the interface designed, the CF platform uses each row in our data set to populate tasks, and links back the necessary data. Each coding task is served up randomly by CF to its pool of workers, and the job runs on the platform until the desired number of trusted judgments has been collected.

Our job settings for each CrowdFlower job are reported in Table 7. Full materials including all of the data files, CML, and instructions required to replicate the data production process on CrowdFlower are provided in the replication materials.

Coding economic and social policy statements in political text (20131210)

Instructions ▲

This task involves reading sentences from political texts and judging whether these deal with economic or social policy.

The sentences you will be asked about come from political party manifestos. Some deal with economic policy, some with social policy; others deal with neither economic nor social policy. We tell you below what we mean by "economic" and "social" policy.

First, you will read a short section from a party manifesto. For the sentence highlighted in red, enter your best judgment about whether it mainly refers to economic policy, to social policy, or to neither.

If the sentence refers to economic policy, select "economic" in the drop down menu; if it refers to social policy, select "social". If the sentence does not refer to either policy area, select "not economic or social" – in this case you will move directly to the next sentence.

If you select "economic" or "social" policy, you will be asked to give your best judgment of the sentence how much it is left or right wing (for economic policy), or liberal or conservative (for social policy). We tell you below about what we mean by "left" and "right" etc.

For example, if you see a sentence with what you think is very right wing economic policy, select the economic policy area from the drop down menu and click "very right" on the economic policy scale. If you think it has a position that is left-wing, but not very left wing, click "left".

If you believe the sentence expresses a centrist position on economic or social policy OR concerns economic or social policy but does not express any clear position, select the appropriate policy category from the drop down menu, as above, and click the "neither…nor…" position on the scale.

Now we will tell you about what we mean by "economic" and "social" policy, and by "left" and "right", "liberal" and "conservative".

What is "economic" policy? What are "left" or "right" economic policies?

"Economic" policies deal with all aspects of the economy, including:

- Taxation and government spending
- Services provided by government or other public bodies
- Pensions, unemployment and welfare benefits; other state benefits
- Property, investment and share ownership, public or private
- Interest rates and exchange rates
- Regulation of economic activity, public or private
- Relations between employers, workers and trade unions

"Left" economic policies tend to favor one or more of the following:

- High levels of state benefits and services provided by the government, even if these imply high levels of taxation;
- Public investment; public ownership or control of sections of business and industry;
- Public regulation of private business and economic activity;
- Support for workers/trade unions relative to employers

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Examples

Below we provide two examples of text from the manifestos and instructions on how they should be coded, and why.

Example 1: "Right" economic policy:

With a Conservative Government, all that has been changing. **We were determined to make share-ownership available to the whole nation.** Just as with cars, television sets, washing machines and foreign holidays, it would no longer be a privilege of the few

The highlighted text should be coded as "economic" policy because it refers to ownership; and as on the "right" because it promotes private ownership.

Example 2: "Liberal" social policy:

ALTERNATIVES TO PRISON. Every effort should be made to ensure that fine defaulters, elderly shoplifters and drunks are not sent to prison. **Police cautions and intermediate treatment should be more widely used.** Where punishment is appropriate, it should normally be community service rather than prison

The highlighted text should be coded as having to do with "social" policy because refers to policing, and as "liberal" because it promotes alternatives to prison.

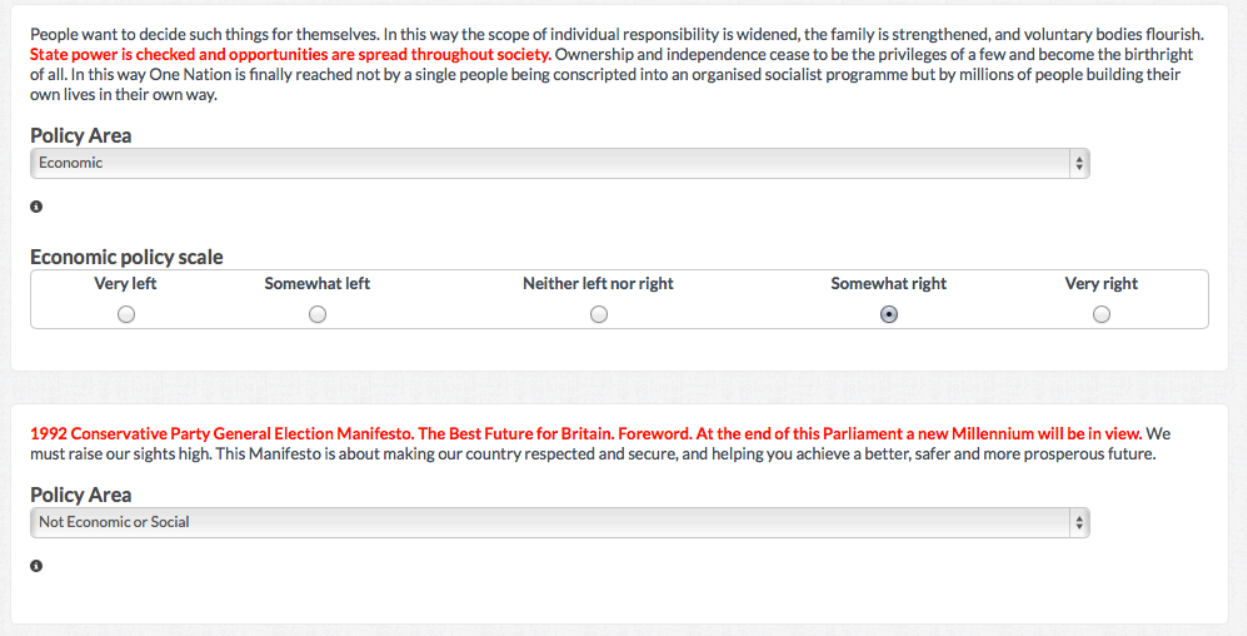


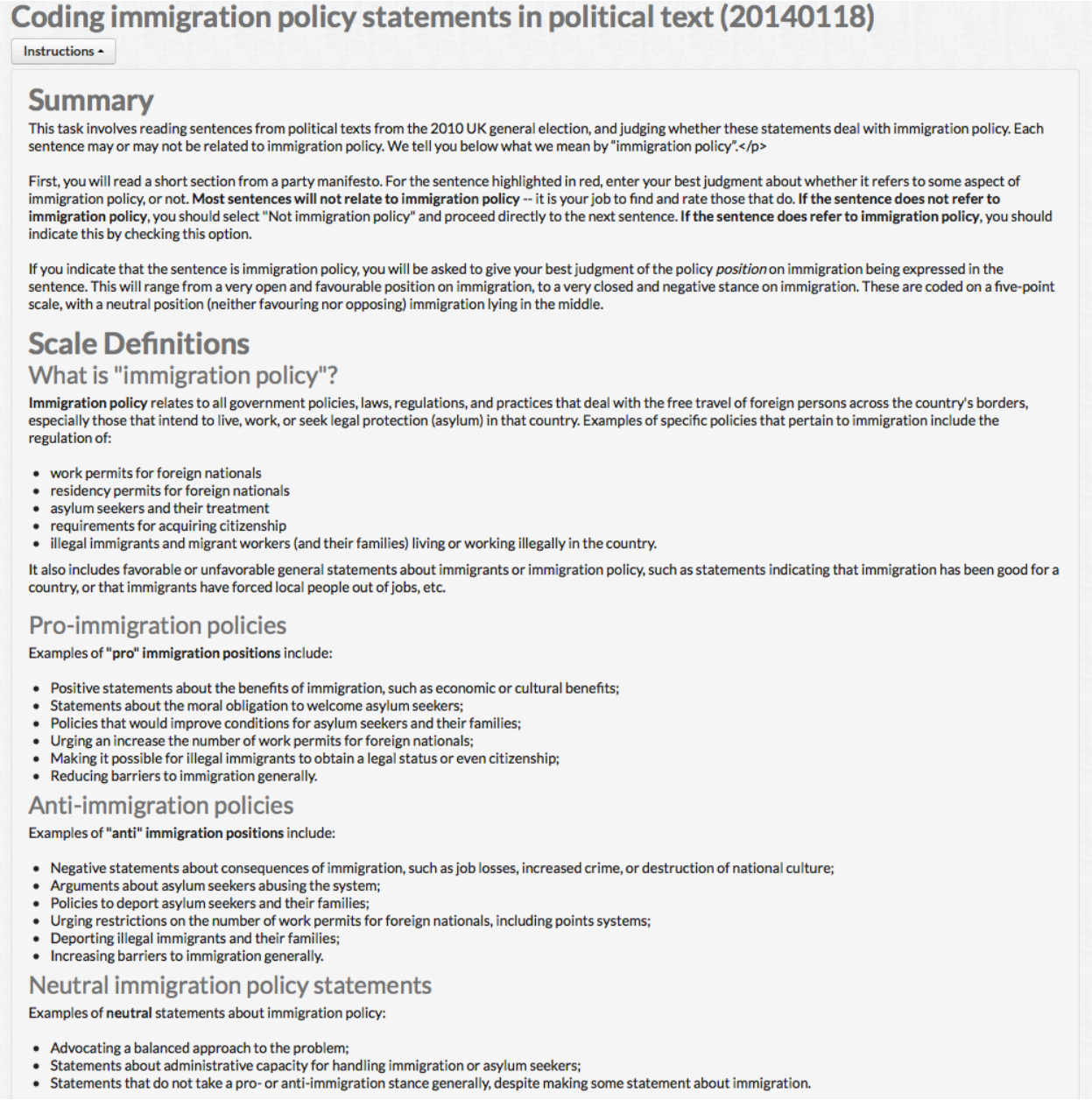
Figure 5. Screenshot of text coding platform, implemented in CrowdFlower.

```
<p>
  {{pre_sentence}}  <strong><font color="red">
  {{sentence_text}}</font></strong>
  {{post_sentence}}</p>
  <cml:select label="Policy Area" class="" instructions="" id=""
validates="required" gold="true" name="policy_area">
    <cml:option label="Not Economic or Social" id="" value="1"></cml:option>
    <cml:option label="Economic" value="2" id=""></cml:option>
    <cml:option label="Social" value="3" id=""></cml:option>
  </cml:select>

   <cml:ratings class="" from="" to="" label="Economic policy scale" points="5"
name="econ_scale" only-if="policy_area:[2]" gold="true" matcher="range">
    <cml:rating label="Very left" value="-2"></cml:rating>
    <cml:rating label="Somewhat left" value="-1"></cml:rating>
    <cml:rating label="Neither left nor right" value="0"></cml:rating>
    <cml:rating label="Somewhat right" value="1"></cml:rating>
    <cml:rating label="Very right" value="2"></cml:rating>
  </cml:ratings>

  <cml:ratings class="" from="" to="" label="Social policy scale" name="soc_scale"
points="5" only-if="policy_area:[3]" gold="true" matcher="range">
    <cml:rating label="Very liberal" value="-2"></cml:rating>
    <cml:rating label="Somewhat liberal" value="-1"></cml:rating>
    <cml:rating label="Neither liberal nor conservative" value="0"></cml:rating>
    <cml:rating label="Somewhat conservative" value="1"></cml:rating>
    <cml:rating label="Very conservative" value="2"></cml:rating>
  </cml:ratings>
```

Figure 6. CrowdFlower Markup Language used for Economic and Social Coding.



Coding immigration policy statements in political text (20140118)

Instructions

Summary

This task involves reading sentences from political texts from the 2010 UK general election, and judging whether these statements deal with immigration policy. Each sentence may or may not be related to immigration policy. We tell you below what we mean by "immigration policy".</p>

First, you will read a short section from a party manifesto. For the sentence highlighted in red, enter your best judgment about whether it refers to some aspect of immigration policy, or not. **Most sentences will not relate to immigration policy** -- it is your job to find and rate those that do. **If the sentence does not refer to immigration policy**, you should select "Not immigration policy" and proceed directly to the next sentence. **If the sentence does refer to immigration policy**, you should indicate this by checking this option.

If you indicate that the sentence is immigration policy, you will be asked to give your best judgment of the policy *position* on immigration being expressed in the sentence. This will range from a very open and favourable position on immigration, to a very closed and negative stance on immigration. These are coded on a five-point scale, with a neutral position (neither favouring nor opposing) immigration lying in the middle.

Scale Definitions

What is "immigration policy"?

Immigration policy relates to all government policies, laws, regulations, and practices that deal with the free travel of foreign persons across the country's borders, especially those that intend to live, work, or seek legal protection (asylum) in that country. Examples of specific policies that pertain to immigration include the regulation of:

- work permits for foreign nationals
- residency permits for foreign nationals
- asylum seekers and their treatment
- requirements for acquiring citizenship
- illegal immigrants and migrant workers (and their families) living or working illegally in the country.

It also includes favorable or unfavorable general statements about immigrants or immigration policy, such as statements indicating that immigration has been good for a country, or that immigrants have forced local people out of jobs, etc.

Pro-immigration policies

Examples of **"pro" immigration positions** include:

- Positive statements about the benefits of immigration, such as economic or cultural benefits;
- Statements about the moral obligation to welcome asylum seekers;
- Policies that would improve conditions for asylum seekers and their families;
- Urging an increase the number of work permits for foreign nationals;
- Making it possible for illegal immigrants to obtain a legal status or even citizenship;
- Reducing barriers to immigration generally.

Anti-immigration policies

Examples of **"anti" immigration positions** include:

- Negative statements about consequences of immigration, such as job losses, increased crime, or destruction of national culture;
- Arguments about asylum seekers abusing the system;
- Policies to deport asylum seekers and their families;
- Urging restrictions on the number of work permits for foreign nationals, including points systems;
- Deporting illegal immigrants and their families;
- Increasing barriers to immigration generally.

Neutral immigration policy statements

Examples of **neutral** statements about immigration policy:

- Advocating a balanced approach to the problem;
- Statements about administrative capacity for handling immigration or asylum seekers;
- Statements that do not take a pro- or anti-immigration stance generally, despite making some statement about immigration.

Figure 7. Immigration Policy Coding Instructions.

```
<p>
  {{pre_sentence}}  <strong><font color="red">
  {{sentence_text}}</font></strong>
  {{post_sentence}}</p>
  <cml:select label="Immigration Policy" class="" instructions="" id=""
validates="required" gold="true" name="policy_area">
    <cml:option label="Not immigration policy" id="" value="1"></cml:option>
    <cml:option label="Immigration policy" value="4" id=""></cml:option>
  </cml:select>

   <cml:ratings class="" from="" to="" label="Immigration policy scale" points="3"
name="immigr_scale" only-if="policy_area:[2]" gold="true">
    <cml:rating label="Favorable and open immigration policy" value="-
1"></cml:rating>
    <cml:rating label="Neutral" value="0"></cml:rating>
    <cml:rating label="Negative and closed immigration policy"
value="1"></cml:rating>
  </cml:ratings>
```

Figure 8. CrowdFlower Markup Language used for Immigration Policy Coding.

7. Instructions for "Coding sentences from a parliamentary debate"

Summary

This task involves reading sentences from a debate over policy in the European parliament, and judging whether particular statements were for or against a proposed policy.

Background

The debates are taken from a debate in the European Parliament over the ending of state support for uncompetitive coal mines.

In general, state aid for national industry in the European Union is not allowed, but exceptions are made for some sectors such as agriculture and energy. At stake here were not only important policy issues as to whether state intervention is preferable to the free market, but also the specific issue for some regions (e.g. Ruhrgebiet in Germany, the north-west of Spain, the Jiu Valley in Romania) where the social and economic impacts of closure would be significant, possibly putting up 100,000 jobs at risk when related industries are considered.

Specifically, the debate concerned a proposal by the European Commission to phase out all state support by 2014. Legislation passed in 2002 that allowed for state subsidies to keep non-profitable coal mines running was due to end in 2010. The Commission proposed to let the subsidies end, but to allow limited state support until 2014 in order to soften the effects of the phase-out. A counter proposal was introduced to extend this support until 2018, although many speakers took the opportunity to express very general positions on the issue of state subsidies and energy policy.

Your Coding Job

Your key task is to judge individual sentences from the debate according to **which of two contrasting positions they supported**:

- **Supporting the rapid phase-out of subsidies for uncompetitive coal mines.** This was the essence of the council proposal, which would have let subsides end while offering limited state aid until 2014 only.

- **Supporting the continuation of subsidies for uncompetitive coal mines.** In the strong form, this involved rejecting the Commission proposal and favoring continuing subsidies indefinitely. In a weaker form, this involved supporting the compromise to extend limited state support until 2018.

Examples of anti-subsidy positions:

- Statements declaring support for the commission position.
- Statements against state aid generally, for reasons that they distort the market.
- Arguments in favor of the Commission phase-out date of 2014, rather than 2018.

Examples of pro-subsidy positions:

- General attacks on the Commission position.
- Arguments in favor of delaying the phase-out to 2018 or beyond.

- Arguments that keeping the coal mines open to provides energy security.
- Arguments that coal mines should be kept open to provide employment and other local economic benefits.
- Preferences for European coal over imported coal, for environmental or safety reasons.

Sample coded Sentences

Below we provide several examples of sentences from the debate, with instructions on how they should be coded, and why.

Example 1: "Anti-subsidy" statement:

The economic return from supporting coal mining jobs through state aid is negative. Furthermore, this money is not being spent on developing sustainable and competitive employment for the future. **Therefore, I believe it is right to phase out state subsidies for uncompetitive mines in 2014.** Instead, we should invest the money into education and training. Only in this way can European remain globally competitive globally.

The highlighted text should be coded as anti-subsidy, because it supports phase-out of subsidies and also specifically supports the Commission deadline of 2014.

Example 2: "Pro-subsidy" statement:

Energy dependency of numerous EU countries, including Spain, puts European security at risk. Losing our capacity to produce our own coal puts Europe at the mercy of foreign suppliers. **This is why state aid to support indigenous production should be maintained.** This would ensure that the EU maintains control over its energy supply rather than depending on foreign coal. It also preserves preservation of thousands of jobs on which significant regions of Europe are largely dependent.

The highlighted text should be coded as pro-subsidy, because it argues that state support should be continued, in the context of both energy security and jobs. It is valid to use the context sentences if the highlighted sentence makes references to them, such as the “This is why…” in the highlighted sentence here.

Example 3: Neutral statements on ending coal subsidies

Thank you Mr. Rapkay for those carefully considered comments. Our fellow Members in this Chamber have mentioned that issue is not new. **It is indeed not new, but is now taking place in different economic and social conditions from before.** We are in a global recession and the European Union is in crisis. No one believes that we have emerged from this crisis yet.

The highlighted text should be coded as neutral because it makes general points not directly related to the Commission proposal or taking a stance on supporting versus ending state subsidies.

Example 4: Test sentences

For several decades, the coal industry has been calling for this transition to be extended, with no end in sight. Equally, for several decades, many European countries have been striving to put an end to what is an unsustainable industry. **Ignore the context and code this sentence as a neutral**

statement on subsidies. We therefore support the Commission's proposal and, by extension, the proposal for subsidies to be used so that the workers concerned can be redeployed in a decent and dignified fashion.

Note that the surrounding sentences may well not match your assessment of the test sentence. However, if you see a sentence like this, please follow its instructions carefully. These sentences are used to check our method and see whether people are paying attention!

CML code:

```
<p>

{{pre_sentence}}
  <strong><font color="red">
{{sentence_text}}</font></strong>

{{post_sentence}}
</p>
<cml:ratings label="On the question of continued subsidies, this statement
is:" points="3" name="subsidy_scale" gold="true" validates="required">
  <cml:rating label="Anti-subsidy" value="-1"></cml:rating>
  <cml:rating label="Neutral or inapplicable" value="0"></cml:rating>
  <cml:rating label="Pro-subsidy" value="1"></cml:rating>
 </cml:ratings>
```

8. Full crowd-sourced coding results for immigration policy

		First Round		Second Round	
Party	Total Sentences in Manifesto	Position	95% CI	Position	95% CI
British National Party	1,240	2.76	[2.33, 3.2]	2.89	[2.33, 3.41]
UK Independence Party	422	2.18	[1.54, 2.9]	2.12	[1.51, 2.78]
Labour	1,349	0.20	[-0.34, 0.69]	0.94	[0.31, 1.58]
Conservative Party	1,240	0.05	[-0.66, 0.86]	0.53	[-0.51, 1.21]
Conservative - LD Coalition	614	0.07	[-0.59, 0.96]	0.19	[-1.11, 1.28]
Plaid Cmyru	339	-1.51	[-2.97, -0.5]	-0.68	[-1.62, 0.37]
Liberal Democrats	855	-1.24	[-2, -0.55]	-0.20	[-0.73, 0.39]
Scottish National Party	369	-0.76	[-2.14, 0.3]	-1.56	[-2.57, 0.6]
Greens	894	-1.48	[-2.28, -0.92]	-1.38	[-2.09, -0.57]
Total	7,322				
Date		Dec. 2013		Feb. 2014	
Total judgments		24,674		24,551	
Correlation with Benoit (2010) Expert Survey		0.96		0.94	
Correlation with Chapel Hill Expert Survey		0.94		0.91	
Correlation with Mean of Means method		0.99		0.98	

Table 10. Crowd-sourced estimates of immigration policy during the 2010 British elections. 95% Bayesian credible intervals in brackets. Expert survey data: Benoit (2010), Marks (2010).

9. Instructions for the crowd-sourced analysis of the first round reviews

Summary

This task involves reading sentences from anonymous reviews of an academic research paper submitted to a political science journal, and judging whether particular statements were for or against publishing the article.

The academic research paper is about whether crowd-sourced coding of sentences (of the type specified in this job) can provide a valid method of determining the overall position expressed in a text.

Each sentence to be coded will be shown in red, surrounded by two sentences of context that you may use in resolving references in the target sentence. You will code them on a five point scale, from strongly supporting publication to strongly supporting rejection.

Your Coding Job

Your key task is to judge individual sentences from the reviews according to **which of two contrasting positions they supported**:

- **Supporting publication of the paper, because it was novel and offered a positive contribution to the field.** This position might have involved revising the paper, but would have expressed its potential in a positive manner.

- **Rejecting that the paper should be published, because it failed to make sufficient scholarly contribution or had too many problems to justify publication.** This position would consist of negative or skeptical statements about the paper, pointing out problems in the logic or evidence in the research, or casting doubt as to its general scholarly worthiness or novelty.

Sample coded Sentences

Below we provide several examples of sentences from each side, with instructions on how they should be coded, and why.

Example 1: Pro-publication statement:

The authors have proven how crowd-sourcing can operate cheaply and quickly to produce valid data about political texts. **This provides an effective demonstration of the potential for this method to transform the way that we do research.** For these reasons I think the manuscript would be of great interest to the rest of the discipline.

The highlighted text should be coded as supporting publication, because it provides a very positive statement about the research and how it would positively change research practice if made public (i.e., published).

Example 2: Anti-publication / pro-rejection statement:

The paper is a competent and thorough demonstration of the benefits of the use of crowd-sourced data to capture political party positions. Overall, this paper offers a convincing demonstration of the robustness of crowd sourcing. **However, the paper does not offer a significant enough**

contribution to justify publication in the APSR. The methods it recommends are neither general nor novel enough to be of widespread interest to general researchers in political science.

The highlighted text should be coded as anti-publication, since despite the preceding positive statement, its overall judgment is that the paper should not be published. (The “APSR” referred to is the *American Political Science Review*, the top academic journal in political science.)

Example 3: Neutral statement.

The authors do note a higher variance for the estimates of social policy, but this seems like quite an important result that should be discussed in greater detail. **In addition to the above, there is one minor point that the authors may wish to clarify.** The authors rely very heavily on supplementary materials found in the appendix.

The highlighted text should be coded as neutral because it simply provides a transition to more concrete points, without taking a specific side on the publication issue. Despite the call for something to be discussed in greater detail in the preceding sentence, you should code each red sentence on its own terms.

Example 4: Test sentences

This article aims at introducing the idea of crowd-sourcing into the data generation process of political science. **Ignore the context and code this sentence as strongly anti-publication.** The article further shows that the positions generated based on crowd sourcing correlate very well with both externally generated measures and with the results of coding from "experts" following the same procedure as "the crowd".

Note that the surrounding sentences may well not match your assessment of the test sentence. However, if you see a sentence like this, please follow its instructions carefully. These sentences are used to check our method and see whether people are paying attention!

CML CODE:

```
<p>
{{pre_sentence}}
<strong><font color="red">{{sentence_text}}</font></strong>
{{post_sentence}}
</p>
<cml:ratings label="On the question of continued subsidies, this statement
is:" points="3" name="pub_scale" gold="true" validates="required">
  <cml:rating label="Anti-publication" value="-1"></cml:rating>
  <cml:rating label="Neutral" value="0"></cml:rating>
  <cml:rating label="Pro-publication" value="1"></cml:rating>
 </cml:ratings>
```